Magnetic Field and Magnetic Forces

DISCLAIMER

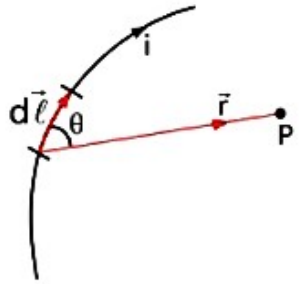
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Biot-Savart Law

Relation between current and magnetic field it produces is given by Biot-Savart law.

The magnetic field at a point P, due to a current element, is given by

$$d\vec{B} = \frac{1}{4\pi\varepsilon_0 c^2}\frac{i\left(d\vec{\ell}\times\vec{r}\right)}{r^3} \qquad ...(i)$$



where, c = speed of light in vacuum ;

i = current in wire ;

$\vec{r}$ = vector joining the current element to the point P ;

= length vector of current element ;

$\frac{1}{\varepsilon_0 c^2} = \mu_0$ = Permeability of vacuum

$\mu_0 = 4\pi \times 10^{-7}$ T m A^{-1} ;

θ = Angle between $d\vec{\ell}$ and $\vec{r}$.

Now equation (i) can be written as

$$d\vec{B} = \frac{\mu_0 i}{4\pi}\frac{\left(d\vec{\ell}\times\vec{r}\right)}{r^3}$$

This equation is the mathematical form of Biot-Savart law.

Note:

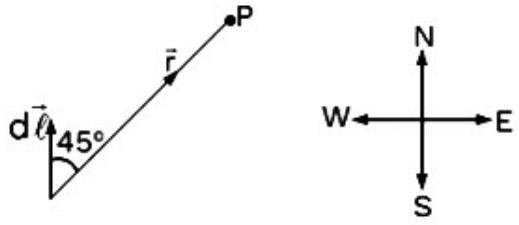
1. The product $\varepsilon\mu$ is related to the speed v of electromagnetic radiation in the medium through $\varepsilon\mu = \frac{1}{v^2}$.
2. Electric permittivity ε is a physical quantity that describes how an electric field affects and is affected by a medium. It is determined by the ability of a material to polarise in response to an applied field and thereby to cancel, partially, the field inside the material.
3. Magnetic permeability (μ) is the ability of a substance to acquire magnetisation in magnetic fields. It is a measure of the exert to which magnetic field can penetrate matter.

Q. **A wire placed along north-south direction carries a current of 10 A from south to north. Find the magnetic field due to a 1 cm piece of wire at a point 200 cm north-east from the piece.**

Sol. Here $1\text{ cm} << 200\text{ cm}$

or $\left|d\vec{\ell}\right| << \left|\vec{r}\right|$

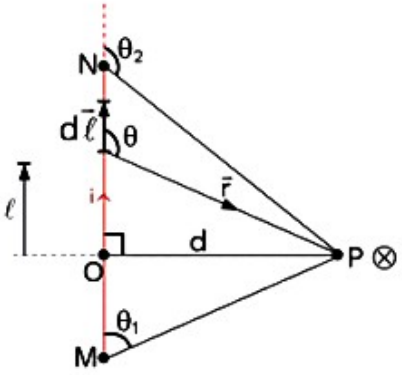
The magnetic field at P is given by-



$$d\vec{B} = \frac{\mu_0}{4\pi}\frac{id\vec{\ell}\times\vec{r}}{r^3} = \frac{\mu_0}{4\pi}\frac{id\ell}{r^2}\sin 45^\circ \qquad d\vec{B} = 10^{-7}\times\frac{10\times10^{-2}}{4}\times 0.707$$

$$d\vec{B} = 1.8\times10^{-9}\text{ T} \qquad \text{(vertically downward)}$$

Magnetic Field Due to Current in a Straight Wire



Let MN be a portion of a straight wire carrying a current i. Let P be a point at a distance OP = d from it. The point O is the foot of the ⊥ from P to the wire.
Let us consider an element $d\ell$ of the wire at a distance ℓ from the point O.
The magnetic field at P due to the element is

$$dB = \frac{\mu_0 i}{4\pi}\frac{d\ell \sin\theta}{r^2} \qquad ...(1)$$

The net field due to the wire MN is obtained by integrating equation (1) under proper limits.
From the figure, $\ell = -d\cot\theta$ and $r = d\operatorname{cosec}\theta$
or $d\ell = +d\,\operatorname{cosec}^2\theta d\theta$
Putting $d\ell$ and r in equation (1),

$$\int dB = \int \frac{\mu_0 i}{4\pi d}\sin\theta d\theta \Rightarrow B = \frac{\mu_0 i}{4\pi d}\int_{\theta_1}^{\theta_2}\sin\theta\, d\theta$$

$$\Rightarrow B = \frac{\mu_0 i}{4\pi d}[-\cos\theta]_{\theta_1}^{\theta_2} \Rightarrow B = \frac{\mu_0 i}{4\pi d}[\cos\theta_1 - \cos\theta_2] \qquad ...(2)$$

Here θ_1 and θ_2 are the value of θ corresponding to the lower end and upper end respectively.

Field on a Perpendicular Bisector:

Suppose the length MN = a and the point P is on its perpendicular bisector. So,

$$OM = ON = \frac{a}{2};$$

$$\cos\theta_1 = \frac{a/2}{\sqrt{\left(\frac{a}{2}\right)^2 + d^2}} \Rightarrow \cos\theta_1 = \frac{a}{\sqrt{a^2 + 4d^2}}$$

and $\cos\theta_2 = \frac{-a}{\sqrt{a^2 + 4d^2}}$

Equation (2) then becomes,

$$B = \frac{\mu_0 i}{4\pi d} \frac{2a}{\sqrt{a^2 + 4d^2}}$$

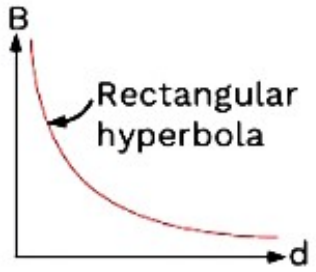
$$\Rightarrow B = \frac{\mu_0 i a}{2\pi d\sqrt{a^2 + 4d^2}} \quad ...(3)$$

Field Due to a Long, Straight Wire:

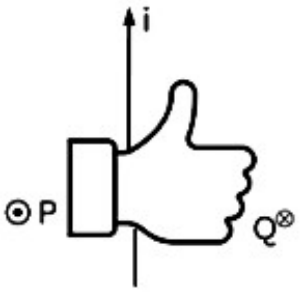
In this case, $\theta_1 = 0$ and $\theta_2 = \pi$. From equation (2), the magnetic field is,

$$B = \frac{\mu_0 i}{2\pi d} \quad ...(4)$$

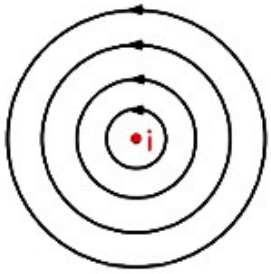
Here $B \propto \frac{1}{d}$, i.e., intensity of magnetic field (crowdedness of field lines) from the wire.

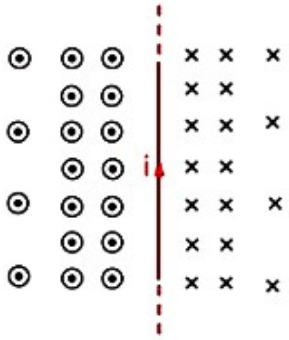


Note: The direction of the magnetic field at a point P due to a long, straight wire can be found by a slight variation in the right-hand thumb rule. If we stretch the thumb of the right hand along the current and curl our fingers to pass through the point P, the direction of the fingers at P gives the direction of the magnetic field there.



In the following way, we can draw the field lines due to a long, straight current



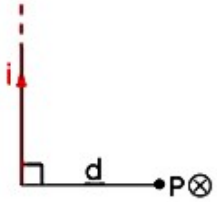


Here, crowdedness of the field lines decreases as we move away from the current source.

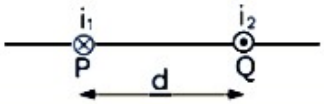
Field Due to Semi-Infinite Straight Wire:

In this case, $\theta_1 = 90°$ and $\theta_2 = 180°$. From equation (2), the magnetic field will be

$$B = \frac{\mu_0 i}{4\pi d} \qquad ...(5)$$



Q. **Two parallel wires P and Q placed at a separation d = 6 cm carry electric currents i_1 = 5A and i_2 = 2A in opposite directions as shown in figures. Find the point on the line PQ where the resultant magnetic field is zero.**



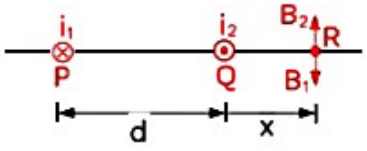
Sol. At point R,

$$\vec{B}_R = \vec{B}_1 + \vec{B}_2 = \vec{O} \qquad \text{or} \qquad |\vec{B}_1| = |\vec{B}_2|$$

$$\Rightarrow \quad \frac{\mu_0 i_1}{2\pi(d+x)} = \frac{\mu_0 i_2}{2\pi x}$$

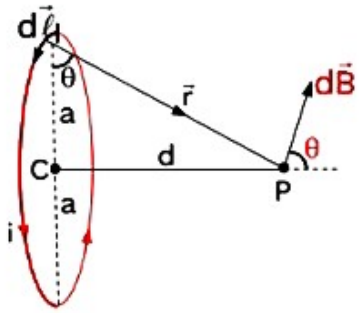
$$\Rightarrow \quad \frac{5}{6+x} = \frac{2}{x} \Rightarrow 5x = 12 + 2x$$

$$\Rightarrow \quad x = 4 \text{ cm}$$



Magnetic Field Due to a Circular Current

1. Field at an Axial Point:



Consider a current element $id\vec{\ell}$ (coming out of the plane) as shown in figure. The magnetic field at P due to this current element is

$$d\vec{B} = \frac{\mu_0 i}{4\pi}\frac{d\vec{\ell}\times\vec{r}}{r^3} \qquad ...(1)$$

As $d\vec{\ell}$ lies $\perp$ to the plane of the figure and $\vec{r}$ lie on the plane of the figure, so we can write equation (1) as

$$dB = \frac{\mu_0}{4\pi}\frac{id\ell r\sin 90^\circ}{r^3} = \frac{\mu_0}{4\pi}\frac{id\ell}{r^2} \Rightarrow dB = \frac{\mu_0}{4\pi}\frac{id\ell}{(a^2+d^2)}$$

The component of field along the axis is

$$B_{axis} = \int dB\cos\theta = \int \frac{\mu_0}{4\pi}\frac{id\ell}{(a^2+d^2)}\cdot\frac{a}{\sqrt{a^2+d^2}}$$

$$B_{axis} = \frac{\mu_0}{4\pi}\frac{ia}{(a^2+d^2)^{3/2}}\int d\ell$$

$$B_{axis} = \frac{\mu_0 ia}{4\pi(a^2+d^2)^{3/2}}.2\pi a$$

$$B_{axis} = \frac{\mu_0 ia^2}{2(a^2+d^2)^{3/2}} \qquad ...(2)$$

The right-hand thumb rule can be used to find the direction of the field. If we curl our right hand fingers in the direction of current, the thumb will indicate the direction of the field.

The component of field $\perp$ to the axis

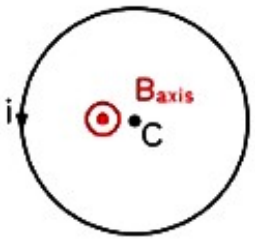
$$\vec{B}_\perp = \int d\vec{B}\sin\theta \Rightarrow B_\perp = 0 \qquad ...(3)$$

The vector sum of all $(d\vec{B}\sin\theta)$ corresponding to all elements will cancel each other.

2. Field at the Centre:

For the central point, d = 0. From equation (2)

$$B_{axis} = \frac{\mu_0 i}{2a} \qquad ...(4)$$



3. Field at a Point far away from the Centre:

Here d >> a and therefore from equation (2)

$$B \approx \frac{\mu_0 i a^2}{2d^3} \Rightarrow B = \frac{\mu_0 i a^2}{2d^3} \times \frac{2\pi}{2\pi}$$

$$\Rightarrow B = \frac{2\mu_0 i(\pi a^2)}{4\pi d^3} \Rightarrow \vec{B} = \frac{2\mu_0 (i\vec{A})}{4\pi d^3}$$

$$\Rightarrow \vec{B} = \frac{\mu_0}{4\pi}\frac{2\vec{\mu}}{d^3} \qquad ...(5)$$

Here $\vec{A} = \pi a^2 \hat{n}$ = Area vector of the loop (and its direction is along the field on the axis);

$\vec{\mu} = i\vec{A}$ = Magnetic dipole moment of the loop.

Note:

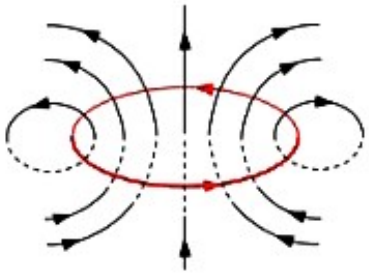
(i) The equation (5) is very similar with the expression for the electric field due to an electric dipole at a far point on the dipole axis. It is

$$\vec{E} = \frac{1}{4\pi\varepsilon_0}\frac{2\vec{P}}{d^3}$$

where $\vec{P}$ = electric dipole moment

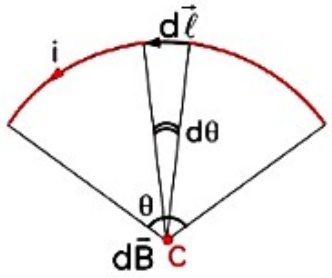
from this similarity, we can now treat a current loop as a magnetic dipole.

(ii) The magnetic field at a point, not on the axis, is mathematically difficult to calculate but magnetic field lines pattern due to a circular loop can give some idea about the intensity and direction of field.



Magnetic Field at the Centre of a Circular Arc

A circular arc of radius 'a', subtends on angle θ at the centre C as shown in figure.



Consider a small element $d\vec{\ell}$ on the arc which subtends on angle dθ at the centre C and produces a small field $d\vec{B}$ at C.

$\because \quad d\vec{B} = \frac{\mu_0 i d\vec{\ell} \times \vec{r}}{4\pi r^3}$ (Biot-Savart Law)

$\therefore \quad dB = \frac{\mu_0 i d\ell}{4\pi a^2} \quad$ or $\quad dB = \frac{\mu_0 i d\theta}{4\pi a} \quad (\text{as } d\ell = ad\theta)$

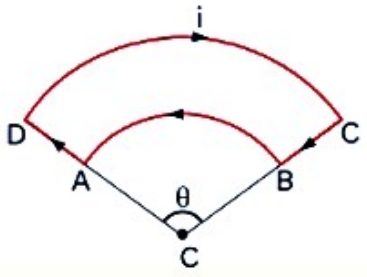
Field at C due to circular arc:

$B = \int dB$

$B = \frac{\mu_0 i}{4\pi a^2}\int d\ell = \frac{\mu_0 i}{4\pi a}\theta$

The direction of the field can be found according to right hand thumb rule.

Q. **Figure shows a current loop having two circular arcs (inner radius 'a' and outer radius 'b') joined by two radial lines (AD and BC). Find the magnetic field at centre C.**



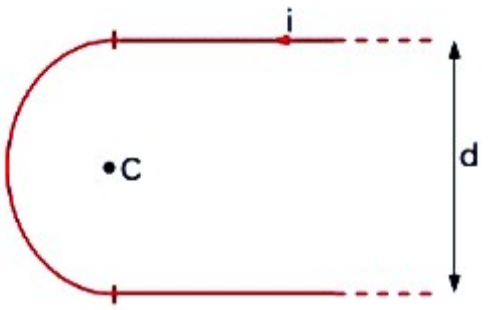
Sol. $\vec{B}_C = \vec{B}_{arc\,CD} + \vec{B}_{arc\,AB} + \vec{B}_{line\,AD} + \vec{B}_{line\,BC}$

$$\vec{B}_C = \frac{\mu_0 i}{4\pi b}\theta \otimes + \frac{\mu_0 i}{4\pi a}\theta \odot + 0 + 0$$

$$\vec{B}_C = \frac{\mu_0 i\theta}{4\pi ab}(b-a)\odot$$

Here line AD and line BC are radial lines therefore their contribution in the field at C will be zero.

Q. **Find the magnetic field at the point C in figure.**



Sol. If $\vec{B}_1$ = field due to upper semi-infinite wire ;

$\vec{B}_2$ = field due to semi-circular wire ;

$\vec{B}_3$ = field due to lower semi-infinite wire ;

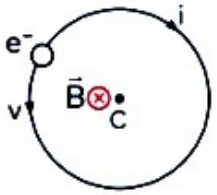
then $\vec{B}_C = \vec{B}_1 + \vec{B}_2 + \vec{B}_3$

$$\vec{B}_C = \frac{\mu_0 i}{4\pi\left(\frac{d}{2}\right)}\odot + \frac{\mu_0 i}{4\pi\left(\frac{d}{2}\right)}(\pi)\odot + \frac{\mu_0 i}{4\pi\left(\frac{d}{2}\right)}\odot$$

$$\vec{B}_C = \left(\frac{\mu_0 i}{\pi d} + \frac{\mu_0 i}{2d}\right)\odot$$

$$\vec{B}_C = \frac{\mu_0 i}{2d}\left(\frac{2}{\pi} + 1\right)\odot$$

Q. **An electron makes 3×10^5 revolutions/sec in circle of radius 0.5Å. Find the magnetic field B at the centre of the circle.**



Sol. The direction of equivalent current will be opposite to the direction of motion of electron (negative charge).

Equivalent current, $i = \frac{q}{T} = \frac{e}{1/f}$

$$i = ef = 1.6 \times 10^{-19} \times 3 \times 10^5$$

$$i = 4.8 \times 10^{-14}\ A$$

Now the field at the centre of the loop is

$$\vec{B}_C = \frac{\mu_0 i}{2a} = \frac{4\pi \times 10^{-7} \times 4.8 \times 10^{-14}}{2 \times 0.5 \times 10^{-10}} \otimes$$

$$B_C = 6.0 \times 10^{-10}\ T$$

Q. **The magnetic field B due to a current carrying circular loop of radius 12 cm at its centre is 0.5×10^{-4} T. Find the magnetic field due to this loop at a point on the axis at a distance of 5 cm from the centre.**

Sol. Field at axial point is

$$B = \frac{\mu_0 i a^2}{2(a^2 + x^2)^{3/2}}$$

At centre,

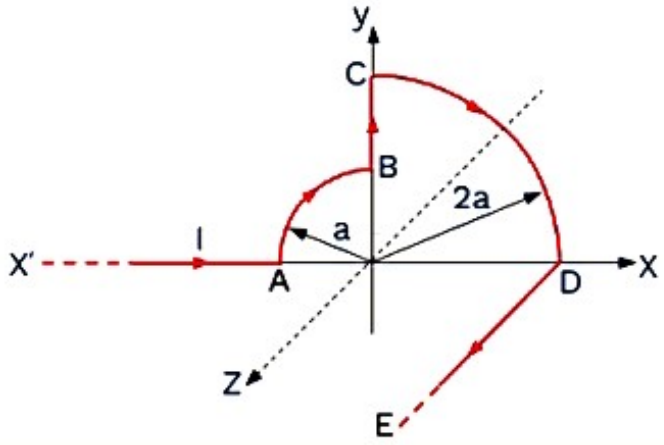
$$B_0 = \frac{\mu_0 i}{2a}$$

$$\because \quad \frac{B}{B_0} = \frac{a^3}{(a^2 + x^2)^{3/2}} \qquad \Rightarrow B = B_0 \frac{a^3}{(a^2 + x^2)^{3/2}}$$

$$\Rightarrow \quad B = (0.5 \times 10^{-4}) \frac{(12\,cm)^3}{(144\,cm^2 + 25\,cm^2)^{3/2}}$$

$$\Rightarrow \quad B = 3.9 \times 10^{-5}\ T$$

Q. **Find the magnetic induction at origin of the co-ordinate system.**



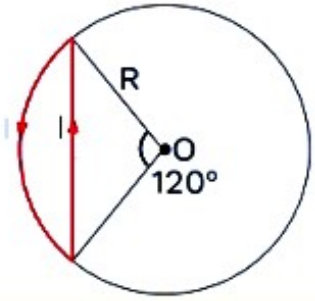
Sol.

$$\vec{B}_0 = \vec{B}_{x'A} + \vec{B}_{AB} + \vec{B}_{BC} + \vec{B}_{CD} + \vec{B}_{DE}$$

$$\vec{B}_0 = 0 + \frac{\mu_0 I}{4\pi a}\left(\frac{\pi}{2}\right)(-\hat{k}) + 0 + \frac{\mu_0 I}{4\pi(2a)}\left(\frac{\pi}{2}\right)(-\hat{k}) + \frac{\mu_0 I}{4\pi(2a)}(-\hat{j})$$

$$\vec{B}_0 = -\frac{\mu_0 i}{8a}\left[\frac{3}{2}\hat{k} + \frac{1}{\pi}\hat{j}\right]$$

Q. **Find magnetic induction at point O in the figure shown due to the current carrying loop with current I as shown.**



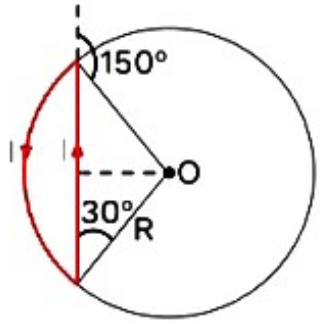
Sol.

$$\vec{B}_0 = \vec{B}_{arc} + \vec{B}_{line}$$

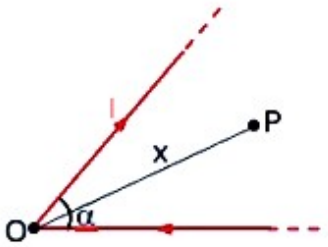
$$\vec{B}_0 = \frac{\mu_0 I}{4\pi R}\left(\frac{2\pi}{3}\right)\odot + \frac{\mu_0 I}{4\pi(R\sin 30^\circ)}[\cos 30^\circ - \cos 150^\circ]\otimes$$

$$\vec{B}_0 = \left(\frac{\mu_0 I}{6R} - \frac{\mu_0 I}{2\pi R}\sqrt{3}\right)\odot$$

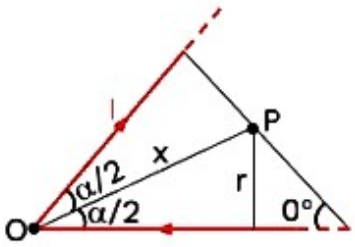
$$\vec{B}_0 = \frac{\mu_0 I}{2R}\left(\frac{1}{3} - \frac{\sqrt{3}}{\pi}\right)\odot$$



Q. **Figure shows a current carrying wire bent at an angle α. Find magnetic induction at a point P located on angle bisector at a distance x from the point O at the bend.**



Sol.



$$B_P = 2\,B_{\text{each segment}}$$

$$= 2\left[\frac{\mu_0 I}{4\pi r}\left(\cos 0^\circ - \cos\left(180^\circ - \frac{\alpha}{2}\right)\right)\right]$$

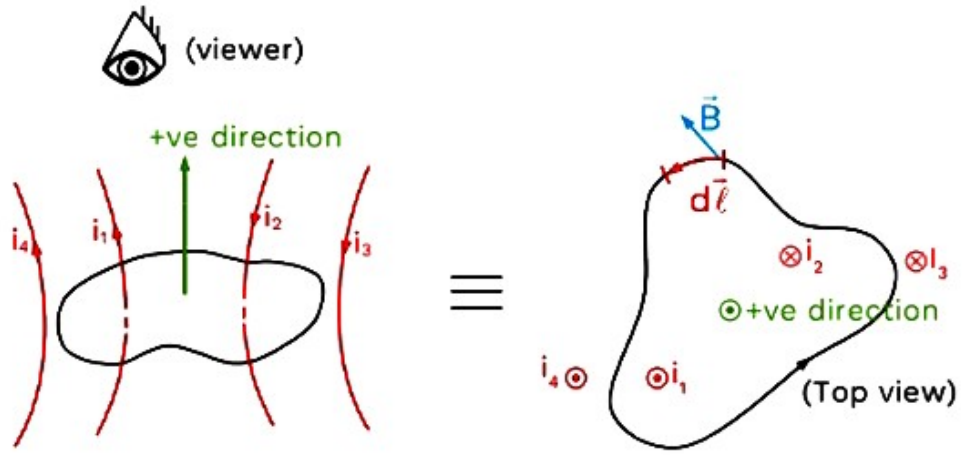
$$= \frac{\mu_0 I}{2\pi\left(x\sin\frac{\alpha}{2}\right)}\left(1+\cos\frac{\alpha}{2}\right)$$

$$= \frac{\mu_0 I\left(2\cos^2\frac{\alpha}{4}\right)}{2\pi\times\left(2\sin\frac{\alpha}{4}\cos\frac{\alpha}{4}\right)}$$

$$\vec{B}_P = \frac{\mu_0 I}{2\pi x}\cot\frac{\alpha}{4}\otimes$$

Ampere's Law

Ampere's law gives another method to calculate the magnetic field due to a given current distribution.



Consider any closed, plane curve as shown in figure. Assign an arrow on the curve. Using right-hand screw rule, assign one side of the plane as +ve and the other as –ve. If the arrow on the loop is anticlockwise, the +ve side is towards the view. If the arrow is clockwise, the +ve side is away from the viewer.

Take a small length-element $d\vec{\ell}$ on the curve and let $\vec{B}$ be the resultant magnetic field at the position of $d\vec{\ell}$. Calculate the scalar product $\vec{B}.d\vec{\ell}$ and integrate by varying $d\vec{\ell}$ on the closed curve. This integration is called "line integral" or "circulation" of $\vec{B}$-along the curve and is represented by the symbol $\oint \vec{B}.d\vec{\ell}$.

Ampere's law then states:

The circulation $\int \vec{B}.d\vec{\ell}$ *of the resultant magnetic field along a closed plane curve is equal to* μ_0 *times the total current* i_{enc} *crossing the area bounded by the closed curve provided the electric field inside the loop remains constant.* Thus,

$$\oint \vec{B}.d\vec{\ell} = \mu_0 i_{enc} \qquad ...(1)$$

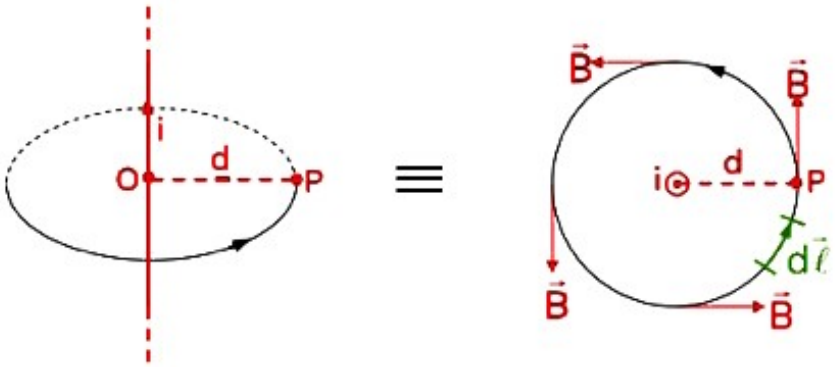
In the above diagram, enclosed or total current is

$$i_{enc} = i_1 - i_2$$

Here, the magnetic field $\vec{B}$ in equation (1) is the resultant field due to all the currents existing anywhere.

Ampere's law may be derived from the Biot-Savart law and Biot-Savart law may be derived from the Ampere's Law. Thus, the two are equivalent in scientific context. However, under certain symmetrical conditions, the mathematics of finding the magnetic field becomes much simpler if we use the Ampere's Law.

Magnetic field due to a straight infinite current carrying wire:



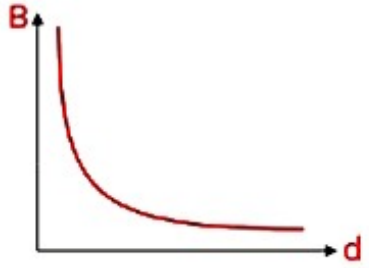
Let us draw a circle passing through the point P and with the centre at O. We put an arrow on the circle according to the right-hand screw rule. The direction of $\vec{B}$ and $d\vec{\ell}$ is always along the tangent at any point on the circle. By symmetry, all points of the circle are equivalent and hence the magnitude of the magnetic field should be the same at all these points. The circulation of magnetic field along the circle is-

$$\oint \vec{B}.d\vec{\ell} = \oint B d\ell = B\oint d\ell = B2\pi d$$

The current crossing the area bounded by the circle is i. Thus, from Ampere's Law:

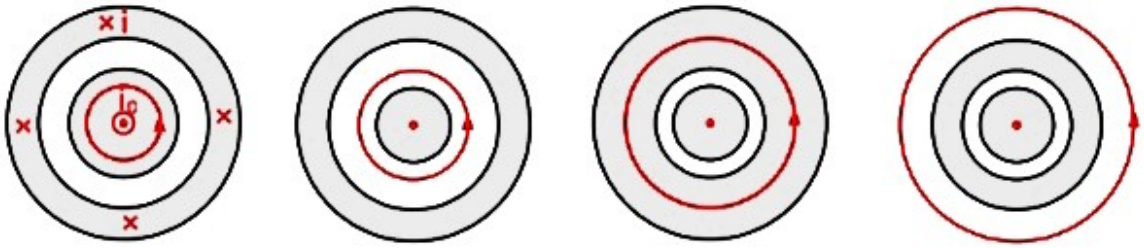
$$B2\pi d = \mu_0 i$$

$$B = \frac{\mu_0 i}{2\pi d}$$



Q. **Consider a coaxial cable which consists of an inner wire of radius 'a' surrounded by a cylindrical shell of inner and outer radii 'b' and 'c' respectively. The inner wire carries an electric current i_0 and outer shell carries an equal current in opposite direction. Find the magnetic field at a distance 'x' from the axis where (a) x < a, (b) a < x < b, (c) B < x < c, (d) x > c.**
Assume that the current density is uniform in the inner wire and also uniform in the outer shell. Also plot the curve between B and x.

Sol.



(a) (b) (c) (d)

A cross-section of the cable is shown in figure. Draw a circle of radius x with the centre at the axis of the cable. The figures (a), (b), (c) and (d) are corresponding to the four parts of the problem. By symmetry, the magnetic field at each point of a circle will have the same magnitude and will be tangential to it. The circulation of B along this circle is, therefore,

$$\oint \vec{B}.d\vec{\ell} = B2\pi x$$

in each of the four parts of the figure.

(a) Apply Ampere's law on given Ampere's loop,

$$\oint \vec{B}.d\vec{\ell} = \mu_0 i_{enc}$$

$$B2\pi x = \frac{\mu_0 i_0 \pi x^2}{\pi a^2} \quad \Rightarrow \quad B = \frac{\mu_0 i_0 x}{2\pi a^2}$$

The direction of field is along the tangent to the circle.

(b) In figure (b),

$$\oint \vec{B}.d\vec{\ell} = \mu_0 i_0$$

$$B2\pi x = \mu_0 i_0 \quad \Rightarrow \quad B = \frac{\mu_0 i_0}{2\pi x}$$

(c) In figure (c),

$$\oint \vec{B}.d\vec{\ell} = \mu_0 \left[i_0 - \frac{i_0(\pi x^2 - \pi b^2)}{(\pi c^2 - \pi b^2)} \right]$$

$$B2\pi x = \frac{\mu_0 i_0 (c^2 - x^2)}{(c^2 - b^2)}$$

$$B = \frac{\mu_0 i_0 (c^2 - x^2)}{2\pi x (c^2 - b^2)}$$

(d) In figure (d),

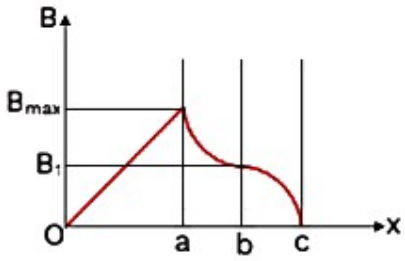
$$\oint \vec{B}.d\vec{\ell} = \mu_0(i_0 - i_0)$$

$$B2\pi x = 0$$

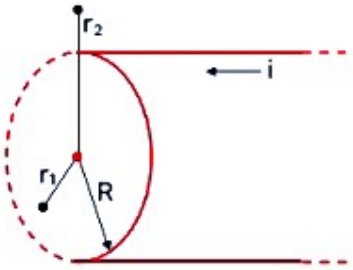
$$B = 0$$

Here $B_{max} = \frac{\mu_0 i_0}{2\pi a}$

$$B_1 = \frac{\mu_0 i_0}{2\pi b}$$



Q. **A long cylindrical conductor of radius R carries a current I as shown in figure. The current density j is a function of radius according to j = br, where b is a constant. Find an expression for the magnetic field B (a) at a distance r_1 < R and (b) at a distance r_2 > R, measured from the axis.**



Sol. (a) Cross-section of cylindrical conductor is shown in figure. Apply Ampere's law on given loop-

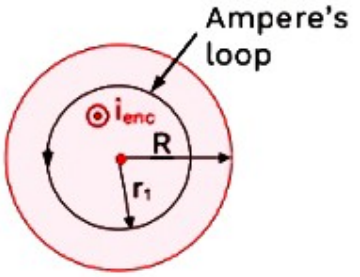
$$\oint \vec{B}.d\vec{\ell} = \mu_0 i_{enc}$$

$$B2\pi r_1 = \mu_0 \int_0^{r_1} j\,2\pi r\,dr$$

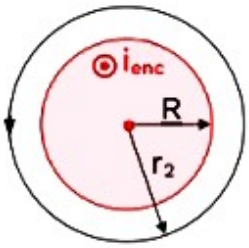
$$Br_1 = \mu_0 \int_0^{r_1} (br)r\,dr$$

$$Br_1 = \mu_0 b \frac{r_1^3}{3}$$

$$\Rightarrow B = \frac{\mu_0 b r_1^2}{3}$$



(b) Apply Ampere's law on given loop-



$$\oint \vec{B}.d\vec{\ell} = \mu_0 i_{enc}$$

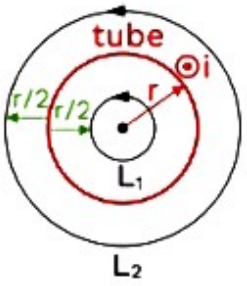
$$B2\pi r_2 = \mu_0 \int_0^R j\,2\pi r\,dr \quad \Rightarrow Br_2 = \mu_0 \int_0^R (br)r\,dr$$

$$B = \frac{\mu_0 b R^3}{3r_2}$$

Q. **A thin but long hollow cylindrical tube of radius r carries a current I along its length. Find the magnitude of the magnetic field at a distance r/2 from the surface (a) inside the tube, (b) outside the tube.**

Sol. The cross-section of tube is shown in figure-

(a) Apply Ampere's law of loop L_1 (inside the tube)



$$\oint \vec{B}.d\vec{\ell} = \mu_0 i_{enc}$$

$$B2\pi\left(r - \frac{r}{2}\right) = \mu_0(0)$$

$$B = 0$$

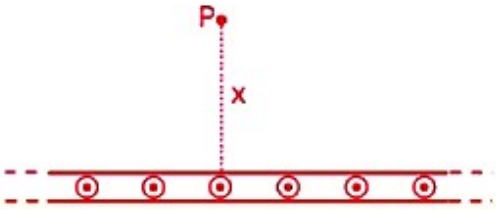
(b) Apply Ampere's law on loop L_2 (outside the loop)

$$\oint \vec{B}.d\vec{\ell} = \mu_0 i_{enc}$$

$$B2\pi\left(r + \frac{r}{2}\right) = \mu_0 i$$

$$B = \frac{\mu_0 i}{3\pi r}$$

Q. **Figure shows a cross-section of a large metal sheet carrying on electric current along its surface. The current in a strip of width $d\ell$ is $Kd\ell$ where K is a constant. Find the magnetic field at a point P at a distance x from the metal sheet.**



Sol.

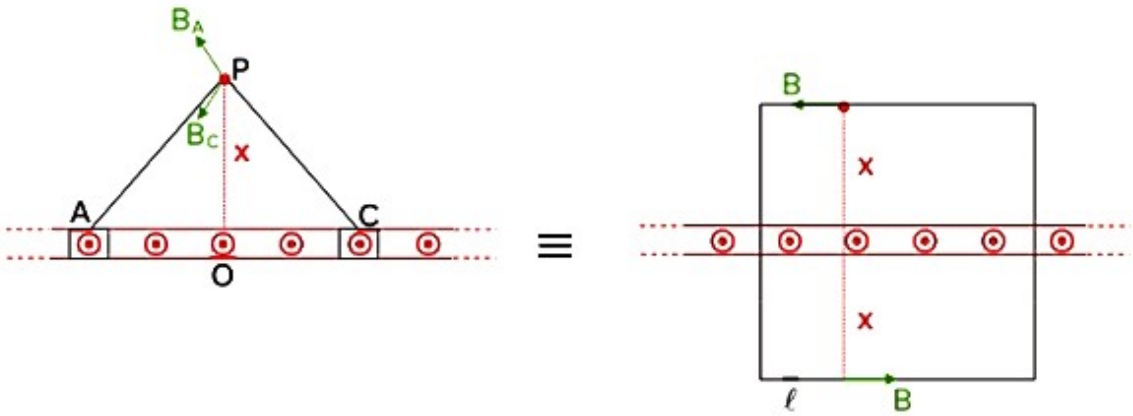


Fig. (a) Fig. (b)

Consider two strips A and C of the sheet situated symmetrically on the two sides of P (figure a). The magnetic field at P due to the strip A is B_a perpendicular to AP and that due to the strip C is B_c perpendicular to CP. The resultant of these two is parallel to width AC. So, the field due to the sheet will also be in this direction.

Apply Ampere's law on the rectangular loop (figure b)-

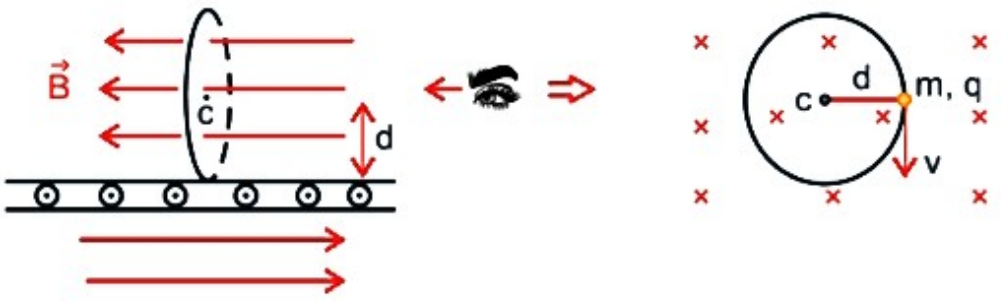
$$\oint \vec{B}.d\vec{\ell} = \mu_0 i_{enc}$$

$$2B\ell = \mu_0(K\ell)$$

$$B = \frac{\mu_0 K}{2} \text{ (field is independent of x)}$$

Q. **Consider the situation described in the previous example. A particle of mass m having a charge q is placed at a distance d from the metal sheet and is projected towards it. Find the maximum velocity of projection for which the particle does not hit the sheet.**

Sol.



Front View Side View

As the field is uniform and the particle is projected in a direction $\perp$ to the field, it will describe a circular path. The particle will not hit the metal sheet if radius $\geq$ d. For the maximum velocity, the radius is just equal to d.
Thus,

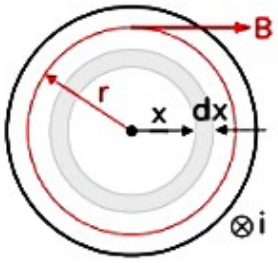
$$q\upsilon B = \frac{m\upsilon^2}{d}$$

$$q\upsilon\left(\frac{\mu_0 K}{2}\right) = \frac{m\upsilon^2}{d}$$

$$\upsilon = \frac{\mu_0 qKd}{2m}$$

Q. **In a long cylindrical wire of radius R, magnetic induction varies with the distance from axis as $B = Cr^{\alpha}$. Find the function of current density in wire with the distance from axis of wire.**

Sol. Apply Ampere's law on Ampere loop-



$$\oint \vec{B}.d\vec{\ell} = \mu_0 I_{enc}$$

$$B\oint d\vec{\ell} = \mu_0 \int_0^r J_x.2\pi x dx$$

$$Cr^{\alpha}2\pi r = \mu_0 \int_0^r J_x 2\pi x dx \Rightarrow Cr^{\alpha+1} = \mu_0 \int_0^r J_x \times dx$$

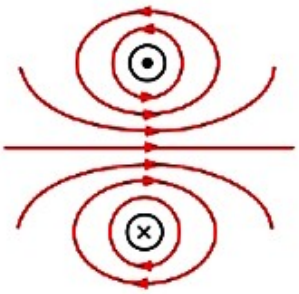
differentiate the equation w.r.t. r

$$C(\alpha + 1)r^{\alpha} = \mu_0 J_r r$$

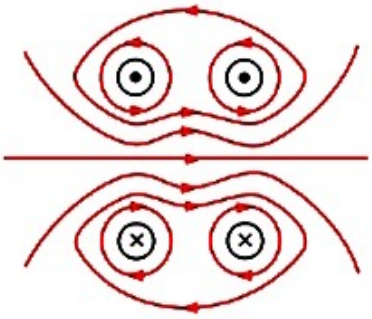
$$J_r = \frac{C(\alpha + 1)r^{\alpha-1}}{\mu_0}$$

Solenoid

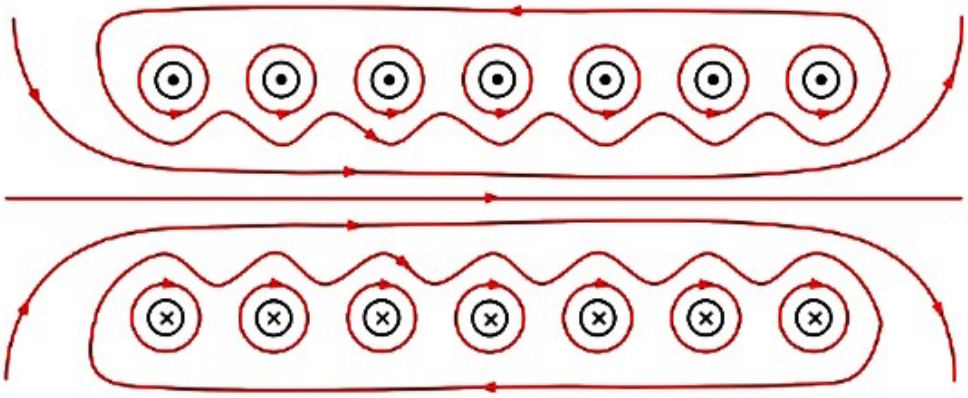
Let us first draw a field line pattern due to single and multi-circular currents (cross-sectional view only)



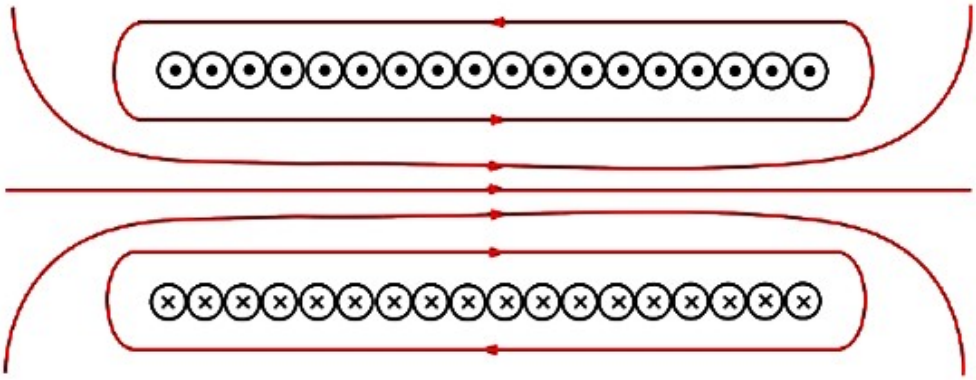
(field due to single loop)



(field due to two loops)

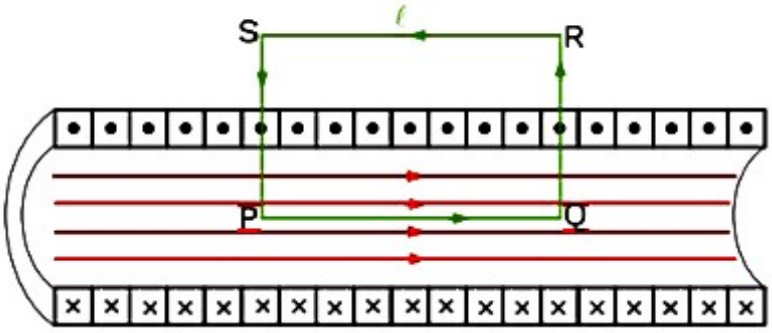


(field due to loosely placed circular loops)



(field due to tightly placed circular loops)

A solenoid is a wire wound closely in the from of a helix. The wire is coated with an insulting material so that although the adjacent turns physically touch each other, they are electrically insulated. Generally, the length of the solenoid is large as compared to the transverse dimension (i.e., radius in case of circular loops/edges in case of rectangular loops).
If we observe the above field line pattern, we can say that a solenoid is like a stack of circular currents placed coaxially one after the other. The fields at an outside point due to the neighbouring loops oppose each other, whereas at an inside point, the fields are in the same direction. These tendencies to have zero field outside and a uniform field inside become more and more effective as the solenoid is more and more tightly wound.
The magnetic field inside a very tightly wound, long solenoid is uniform everywhere and is zero outside it.



To calculate the magnetic field at a point P inside the solenoid, let us draw a Ampere's loop PQRS as shown in figure. The circulation of $\vec{B}$ inside the solenoid is

$$\oint_{PQRS} \vec{B}.d\vec{\ell} = \int_{P}^{Q} \vec{B}.d\vec{\ell} + \int_{Q}^{R} \vec{B}.d\vec{\ell} + \int_{R}^{S} \vec{B}.d\vec{\ell} + \int_{S}^{P} \vec{B}.d\vec{\ell}$$

$$\oint_{PQRS} \vec{B}.d\vec{\ell} = B\ell + \int_{Q}^{R} Bd\ell\cos 90^\circ + \int_{R}^{S} \vec{0}.d\vec{\ell} + \int_{S}^{P} Bd\ell\cos 90^\circ$$

$$\oint_{PQRS} \vec{B}.d\vec{\ell} = B\ell + 0 + 0 + 0$$

$$\oint_{PQRS} \vec{B}.d\vec{\ell} = B\ell$$

If n = number of turns/length of solenoid ;
i = current in each turn ;

then by Ampere's Law

$$\oint_{PQRS} \vec{B}.d\vec{\ell} = \mu_0 i_{enc} = \mu_0(n\ell i)$$

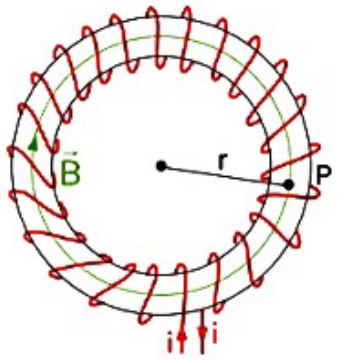
$$B\ell = \mu_0 n\ell i$$

$$B = \mu_0 ni \quad \ldots(1)$$

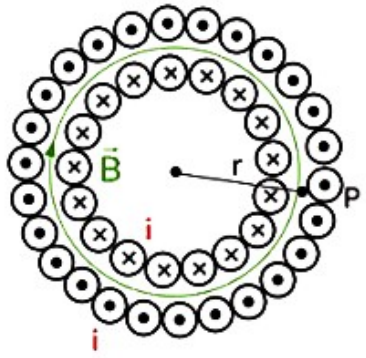
Equation (1) is strictly valid only for an infinitely long solenoid. Practically, if the length of solenoid is more than five to six times of diameter and we are looking at the field in the middle region, we can use this equation.

Toroid

If a solenoid is bent in a circular shape and the ends are joined, we get a toroid. For designing a toroid, one can start with a non-conducting thin tube and wind a conducting wire (with electric insulation) closely on it. The magnetic field in such a toroid can be obtained by using Ampere's Law.



(toroid)



(cross-sectional view of toroid)

If the total number of turns is N, the current crossing the area bounded by the circle is Ni, where i is the current in the toroid. Apply Ampere's law on given Ampere's loop-

$$\oint \vec{B}.d\vec{\ell} = \mu_0 i_{enc}$$

$$\oint B d\ell \cos 0° = \mu_0 (Ni)$$

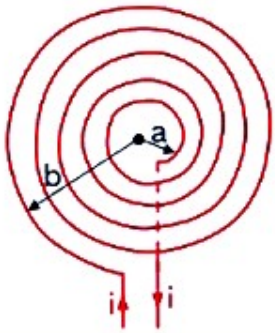
$$B\int d\ell = \mu_0 Ni$$

$$B(2\pi r) = \mu_0 Ni$$

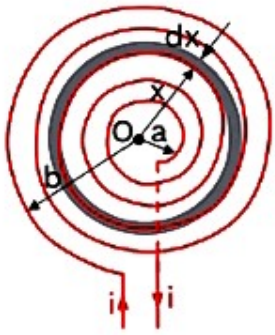
$$B = \frac{\mu_0 Ni}{2\pi r} \quad \text{or} \quad B = \mu_0 ni$$

where $n = \frac{N}{2\pi r}$ = number of turns per unit tangential length.

Q. **Figure shows a spiral of inner radius 'a' and outer radius 'b' with total N turns wound on it. If a current I flows in it, find magnetic induction at the centre of spiral.**



Sol.



Number of turns per unit radial length = $\frac{N}{b-a}$

Number of turns in an elemental coil of radius x and width dx is,

$$dN = \frac{N}{b-a}dx.$$

Magnetic induction due to this elemental coil is-

$$dB = \left(\frac{N}{b-a}\right)dx\ \frac{\mu_0 I}{2x}$$

Total magnetic induction at O is-

$$B = \int dB = \frac{\mu_0 IN}{2(b-a)}\int_a^b \frac{dx}{x}$$

$$B = \frac{\mu_0 IN}{2(b-a)}\ell n\left(\frac{b}{a}\right)$$

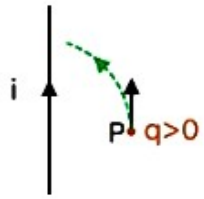
Introduction to Magnetic Field

Consider a wire carrying a current i. Consider a charged particle having charge q placed as shown in figure.



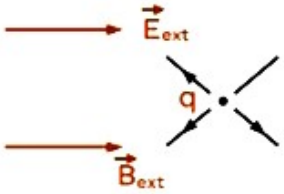
If speed of the charged particle = 0, then force on particle = 0. This is expected because in any volume of wire, there is no net charge, i.e. the wire is electrically neutral and does not produce an electric field.

Now, if we project the charge q along the current, it is deflected towards the wire (for q > 0). We conclude that there must be a field at P which exerts a force on the charge when it is projected, but not when it is kept at rest. This new field is called magnetic field $(\vec{B})$ or magnetic induction and the force exerted by a magnetic field is called "magnetic force $(\vec{F})$".



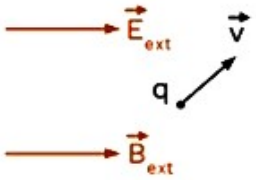
Some facts about electromagnetic Interactions:

1. A static charge produces only electric field and therefore only a external electric field can exert a force on a static charge. The presence of external magnetic field is irrelevant as it can't exert any force on static charge.



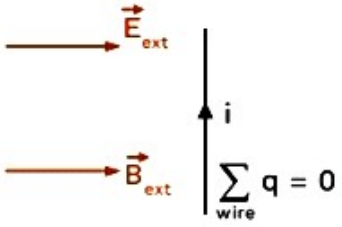
$$\vec{F}_q = f(q, \vec{E}_{ext})$$

2. A moving charge produces both electric and magnetic fields in its surrounding and therefore both external electric and magnetic field can exert force on the moving charge.



$$\vec{F}_q = f(q, \vec{v}, \vec{E}_{ext}, \vec{B}_{ext})$$

3. An electrically neutral current carrying wire produces only magnetic field in its surrounding and therefore only external magnetic field can exert a force on the current carrying conductor.



$$\vec{F}_{wire} = f(i, \vec{\ell}_{wire}, \vec{B}_{ext})$$

Definition of Magnetic Field

If a charged particle is projected in a magnetic field $\vec{B}$, in general, it experiences a magnetic force. By projecting the particle in different directions from the same point P with different speeds, following facts are observed about magnetic force:

(A) There is one line through the point P, such that, if the velocity of the particle is along this line, there is no magnetic force. We define the direction of the magnetic field to be along this line (the direction is not uniquely defined yet, because there are two opposite directions along any line).

(B) If the speed of the particle is $\vec{v}$ and it makes an angle θ with the line identified in (A), i.e., with the direction of the magnetic field, then

$$F \propto |\vec{v} \sin\theta|$$

(C) $\vec{F} \perp \vec{B}$, and $\vec{F} \perp \vec{v}$.

(D) $F \propto q$ and direction of $\vec{F}$ is opposite for positive and negative charges.

All the above facts may be explained if we defined the magnetic field by the equation –

$$\boxed{\vec{F} = q(\vec{v} \times \vec{B})}$$

SI unit of B is-
Newton/ampere-metre = tesla (T) = weber/m^2
Other smaller unit is "Gauss (G)" and 1 T = 10^4G
We have a magnetic field of the order of 10^{-5}T near the earth's surface. The magnets in use today in MRI are in the 0.5T to 3T range while large superconducting magnets are needed to produce a field of the order of 10T in laboratories.

Electromagnetic Force on a Moving Charge in External Fields

Consider a space where an external electric field $\vec{E}$ and an external magnetic field $\vec{B}$ exists, then electromagnetic force $\vec{F}$ experienced by a charge q moving with velocity $\vec{v}$ is given as :

$$\vec{F}_{EM} = \vec{F}_E + \vec{F}_M$$

$$\vec{F}_{EM} = q\vec{E} + q(\vec{v} \times \vec{B})$$

$$\boxed{\vec{F}_{EM} = q(\vec{E} + \vec{v} \times \vec{B})}$$

This is called Lorentz Force equation which give total electromagnetic force and it is constant when observed from any inertial frame of reference.
If we observe from different inertial frame, all parameters $\vec{v}, \vec{E}$ and $\vec{B}$ changes in such a way that individual forces $\vec{F}_E$ and $\vec{F}_M$ may change but net force $\vec{F}_{EM}$ will remain the same.

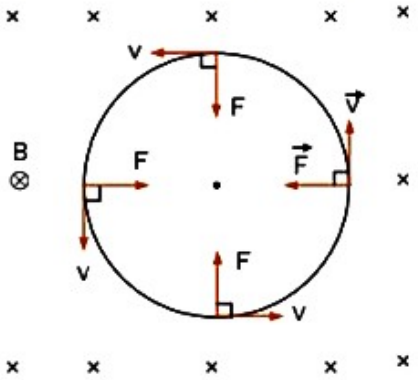
Motion of a Charged Particle in a Uniform Magnetic Field

Case -1: When $\vec{v}$ is $\perp$ to $\vec{B}$,

$\because \vec{F} \perp \vec{v}$

$\therefore P = \vec{F}.\vec{v} = 0$

i.e. magnetic force does not do any work on the particle.

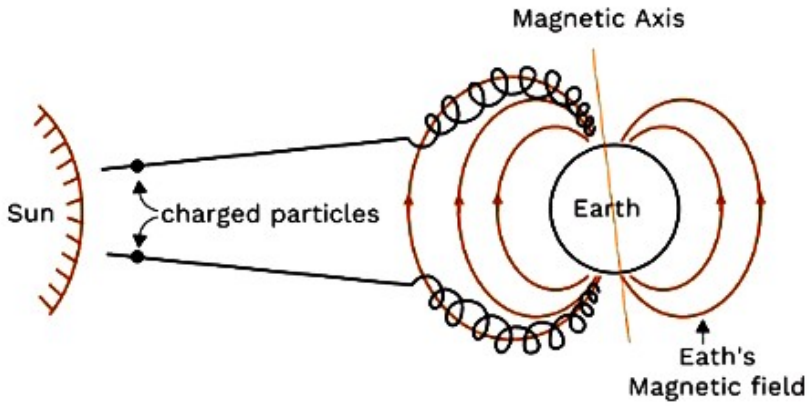


Hence, the kinetic energy or the speed of the particle does not change due to the magnetic force.

The magnetic force will deflect the particle continuously without changing the speed and the particle will move along a circle perpendicular to the field.

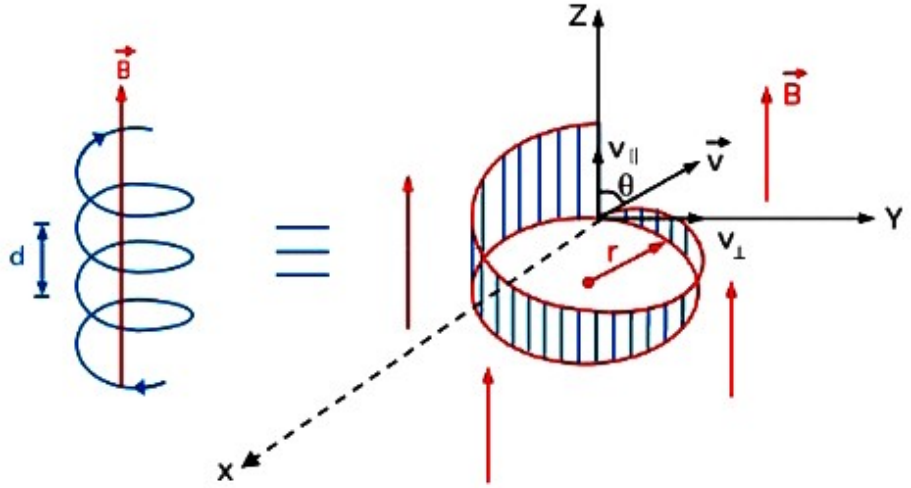
The magnetic force will be reflected in centripetal force. If r be the radius of the circle

$$qvB = \frac{mv^2}{r} \Rightarrow \boxed{r = \frac{mv}{qB}}$$

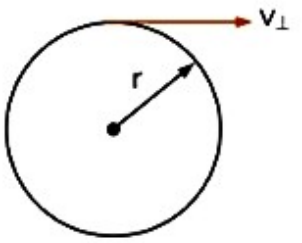


During a solar flare, a large number of electrons and protons are ejected from the sun. Some of them get trapped in the Earth's magnetic field and move in helical paths along the field lines. The field lines come closer to each-other near the magnetic poles (as shown in figure). Hence, the density of charges increases near the poles. These particles collide with atoms and molecules of the atmosphere. Excited oxygen atoms emit green light and excited nitrogen atoms emit pink light. This phenomenon is called "Aurora Borealis".

Case -2: When $\vec{v}$ is not $\perp$ to $\vec{B}$:



$$r = \frac{mv_{\perp}}{qB} = \frac{mv\sin\theta}{qB}$$



Pitch, $d = v_{\parallel}T$

$$d = \frac{2\pi mv_{\parallel}}{qB} = \frac{2\pi mv\cos\theta}{qB}$$

Helical Motion of charged particles and Aurora Borealis:

In polar regions like Alaska, Northern Canada and Antarctica, a splendid display of colours is seen in the sky. The appearance of dancing green pink lights is fascinating, and equally puzzling. An explanation of this natural phenomenon can be explained as following.

The time taken to complete the circle (time period) is:-

$$T = \frac{2\pi r}{v} = \frac{2\pi m}{qB}$$

As v increases, r also increases in same way so that the ratio $\frac{r}{v}$ remains constant so that T is independent of the speed v. In other words, if the particle moves faster, the radius is larger, it has to move a longer distance to complete the circle so that the time taken is the same.

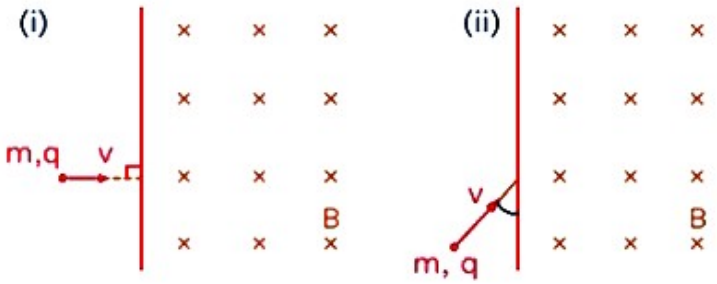
The frequency (cyclotron frequency) of revolution is

$$\nu = \frac{1}{T} = \frac{qB}{2\pi m}$$

and the angular frequency is –

$$\omega = 2\pi\nu \quad \Rightarrow \omega = \frac{qB}{m}$$

Q. **In the following cases, find the radius of curvature, angle of deviation and time spent in the magnetic field:**



Sol. (i) The radius of curvature is-

$$R = \frac{mv}{qB} (\text{as } \vec{v} \perp \vec{B})$$

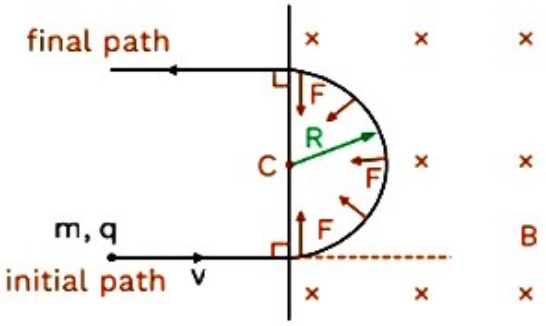
Angle of deviation,

$$\delta = \pi$$

Time spent t is –

$$t = \frac{\text{Path travelled in field}}{\text{Speed of Particle}}$$

$$t = \frac{\pi R}{v} = \frac{\pi R}{(qRB/m)} \Rightarrow t = \frac{\pi m}{qB}$$



(ii) Radius of curvature –

$$R = \frac{mv}{qB} \qquad (\text{as } \vec{v} \perp \vec{B})$$

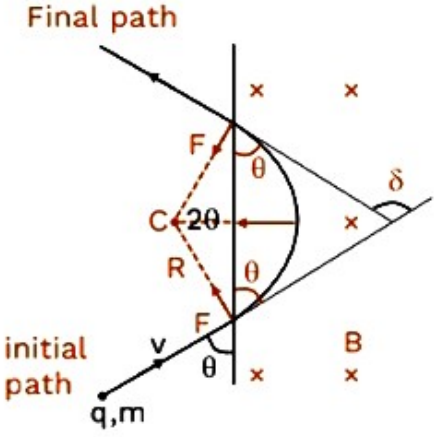
Angle of deviation (δ) -

$\delta = 2\theta$ (from the triangle formed)

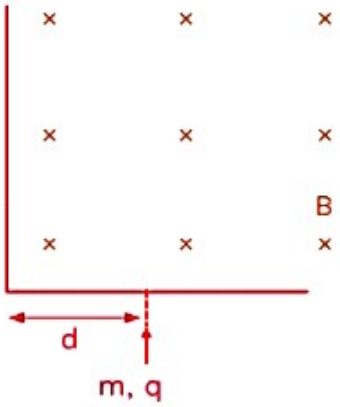
Time spent t is –

$$t = \frac{2\theta}{2\pi} T$$

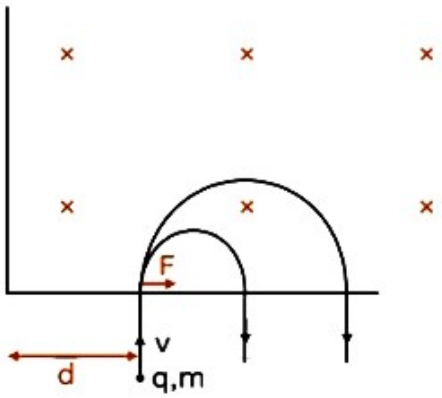
$$t = \frac{2\theta m}{qB}$$



Q. **Find the angle of deviation of charged particle with all possibilities.**

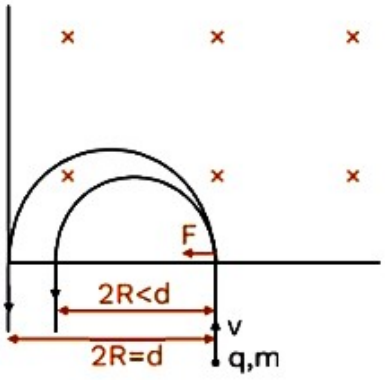


Sol. **Case -1:** If $q < 0$ i.e. the charge is negative,
In this case, the path covered by the particle will be semi-circle and therefore angle of deviation will be π for any value of v.



Case -2 : If $q > 0$ and $2R \leq d$ or if $q > 0$ and $\frac{2mv}{qB} \leq d$

In this case, the path covered by the particle will be semi-circle and therefore angle of deviation will be π for $v \leq \frac{dqB}{2m}$

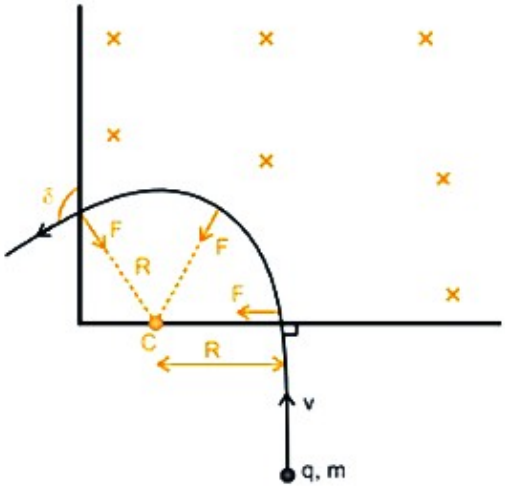


Case -3: If $q > 0$ and $\frac{d}{2} < R < d$

or if $q > 0$ and $\frac{d}{2} < \frac{mv}{qB} < d$

In this case, the angle of deviation δ will be-

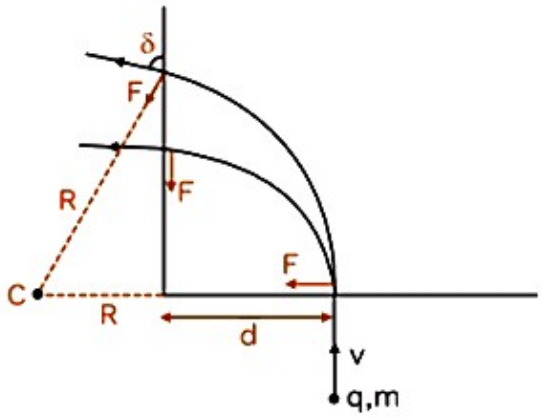
$$\frac{\pi}{2} < \delta < \pi$$



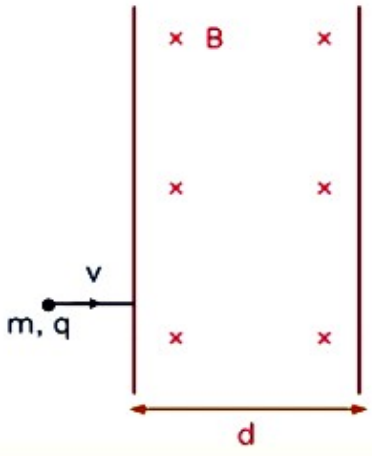
Case -4: If $q > 0$ and $d \leq R < \infty$ or $q > 0$ and $d \leq \frac{mv}{qB} < \infty$

In this case, the angle of deviation δ will be–

$$0 < \delta \leq \frac{\pi}{2}$$



Q. Discuss the angle of deviation of moving charged particle by a given sector of magnetic field.

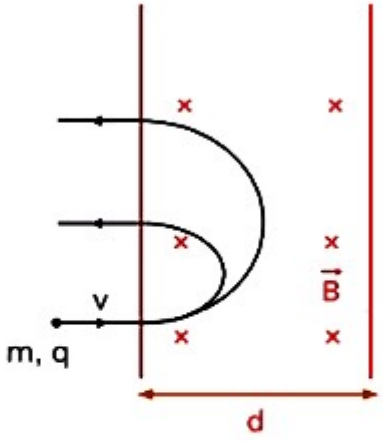


Sol. **Case -1:**

If $R = \frac{mv}{qB} < d$

In this case deviation

$\boxed{\delta = \pi}$



Case -2:

If $R = \frac{mv}{qB} > d$

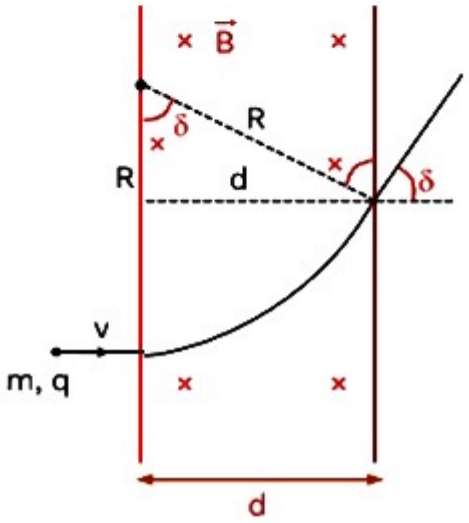
$\because \sin\delta = \frac{d}{R}$

$\therefore \delta = \sin^{-1}\left(\frac{d}{R}\right)$

or $\delta = \sin^{-1}\left(\frac{dqB}{mv}\right)$

From both cases, the set of δ will be-

$\delta \in \left(0, \frac{\pi}{2}\right) + \{\pi\}$



Q. **A charge $4\mu C$ enters in a region of uniform magnetic field with a velocity $\left(4\hat{i} + 7\hat{j}\right)$ m / s experiences a force $\left(5\hat{i} - C\hat{j}\right)$N. Find the value of C.**

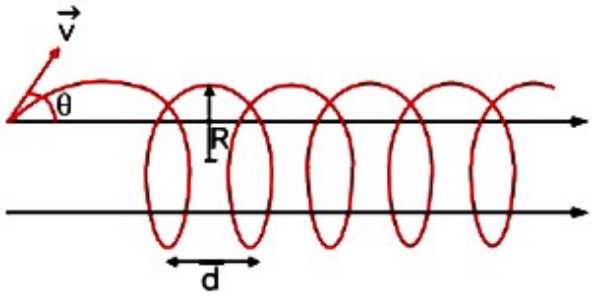
Sol. $\because \vec{F} = q\left(\vec{v} \times \vec{B}\right)$

$\therefore \vec{F}$ is always $\perp$ to $\vec{v} \Rightarrow \vec{F}.\vec{v} = 0$

$\Rightarrow \left(5\hat{i} - C\hat{j}\right).\left(4\hat{i} + 7\hat{j}\right) = 0 \Rightarrow 20 - 7C = 0 \Rightarrow \boxed{C = \frac{20}{7}}$

Q. **A beam of protons with a velocity 8×10^5m/s enters a region of uniform magnetic field of 0.3T at an angle 60° to the direction of magnetic field. Find the radius and pitch of helical path followed by particle.**

Sol.



Radius of helical path,

$$R = \frac{mv\sin\theta}{qB} = \frac{1.67\times10^{-27}\times8\times10^{5}\times(\sqrt{3}/2)}{1.6\times10^{-19}\times0.3}$$

$$R = 2.4\times10^{-2}\,m$$

Pitch of helical path,

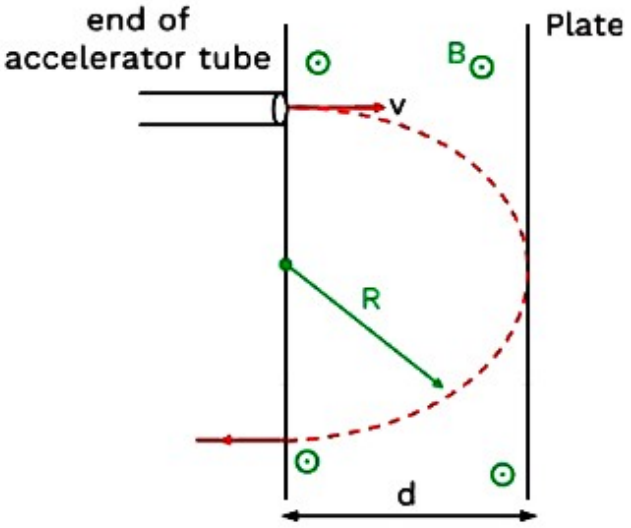
$$d = (v\cos\theta)\,T = (v\cos\theta)\frac{2\pi m}{qB}$$

$$d = \frac{8\times10^{5}\times0.5\times2\times3.14\times1.67\times10^{-27}}{1.6\times10^{-19}\times0.3}$$

$$d = 8.74\times10^{-2}\,m$$

Q. **A beam of singly ionized charged particles, having kinetic energy 1000eV contain particles of masses 9×10^{-27} kg and 1.6×10^{-26} kg energy from one end of an accelerator tube. There is a plate at a distance of 0.01m from end of tube placed perpendicular to it. Find minimum magnetic field in region so that beam will not strike to the plate.**

Sol.



Radius of particles,

$$R = \frac{mv}{qB} = \frac{\sqrt{2mK}}{qB}$$

or $R \propto \sqrt{m}$, i.e. larger radius is for heavy particles.

In order to avoid the striking with the plate,

$$R_{heavy} < d$$

$$\frac{\sqrt{2m_{heavy}K}}{qB} < d \quad \Rightarrow \quad B > \frac{\sqrt{2m_{heavy}K}}{qd}$$

$$\Rightarrow B_{min} = \frac{\sqrt{2 \times 1.6 \times 10^{-26} \times 1000 \times 1.6 \times 10^{-19}}}{1.6 \times 10^{-19} \times 0.01} \Rightarrow \boxed{B_{min} = \sqrt{2}T}$$

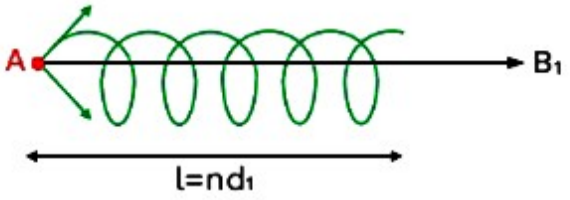
Q. **A slightly divergent beam of charged particles, accelerated by a potential difference V propagates from a point A along the direction of field. The beam is focussed at a point at a distance l from A by two successive magnetic induction value B_1 and B_2 one after another. Find specific charge q/m of the particles.**

Sol. For $B_1 \rightarrow$

$$\ell = nd_1 = n\left(T_1 V \cos\theta\right)$$

$$\ell \approx \frac{n2\pi nV}{qB_1} = \frac{n_2\pi\sqrt{2mqV}}{qB_1}$$

$$\ell B_1 = \frac{n2\pi}{q}\sqrt{2mqV} \qquad ...(1)$$

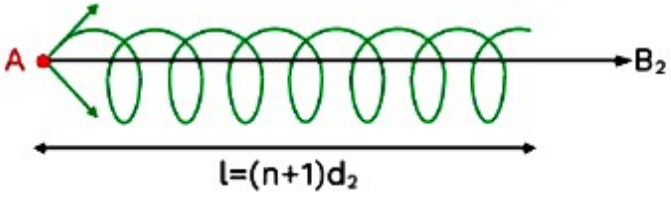


for $B_2 \rightarrow$

$$\ell = (n+1)d_2 = (n+1)\left(T_2 V \cos\theta\right)$$

$$\ell \approx (n+1)\frac{2\pi mV}{qB_2}$$

$$\ell B_2 = \frac{(n+1)2\pi}{q}\sqrt{2mqV} \qquad ...(2)$$



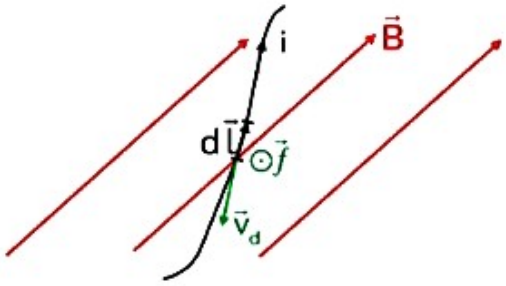
from (1) and (2),

$$(B_2 - B_1)\ell = \frac{2\pi}{q}\sqrt{2mqV} \Rightarrow (B_2 - B_1)^2 \ell^2 = \frac{4\pi^2}{q^2}(2mqV)$$

$$\boxed{\Rightarrow \frac{q}{m} = \frac{8\pi^2 V}{\ell^2 (B_2 - B_1)^2}}$$

Magnetic Force on a Current Carrying Wire

Suppose a conducting wire, carrying a current I is placed in a magnetic field $\vec{B}$. Now focus a length element $d\vec{l}$ of it. The average force on a free e^- present in $d\vec{l}$ length of wire is



$$\vec{f} = (-e)\left(\vec{v}_d \times \vec{B}\right)$$

where $\vec{v}_d$ = drift velocity of free electron (opposite to $d\vec{l}$).

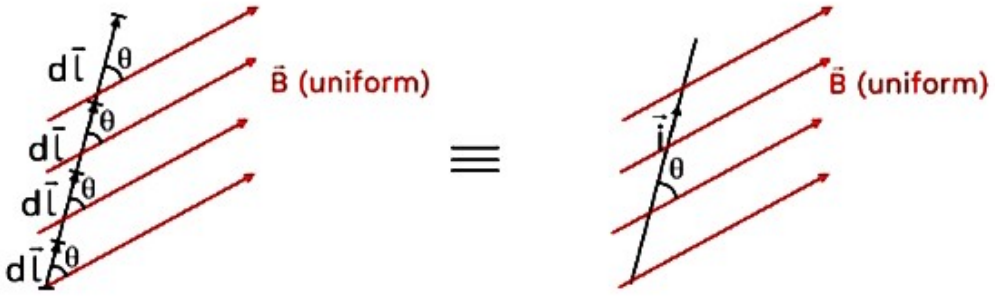
If A is the area of cross-section of the wire and n is the no. of free electrons per unit volume, then total no. of free electrons in the small length element is (n A dl) and magnetic force on it is –

$$d\vec{F} = (-nAdl)(-e\,\vec{v}_d \times \vec{B})$$

$$d\vec{F} = nAev_d\,(e).(d\vec{l} \times \vec{B})$$

$$\boxed{d\vec{F} = i\left(d\vec{l} \times \vec{B}\right)}$$

The quantity $id\vec{l}$ is called current element and the direction of $d\vec{l}$ is taken along the direction of current i.



Now consider a straight wire of length l carrying a current i is placed in a uniform magnetic field $\vec{B}$, then force on it is,

$$\vec{F} = \int d\vec{F} = i\int (d\vec{l} \times \vec{B})$$

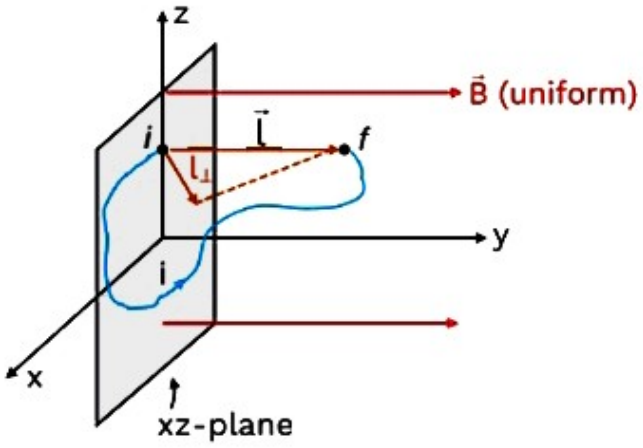
Here, $\vec{B}$ is same at each location of length element $d\vec{l}$. Also, the angle between each length

element $d\vec{l}$ and $\vec{B}$ is same. Therefore, we can write–

$$\vec{F} = i\left(\int d\vec{l}\right) \times \vec{B}$$

$$\boxed{\vec{F} = i\left(\vec{l} \times \vec{B}\right)}$$

Now consider a random shaped current carrying wire placed in a uniform field in such a way that some of its portion lying on xz-plane and other portion out of the plane.



Since, $\vec{B}$ is same at each location of length element $d\vec{l}$. Therefore, net force on wire can be written as–

$$\vec{F} = i\int (dl)B \sin\theta \hat{n} = iB\int (dl \sin\theta \hat{n})$$

Here $\int (dl \sin\theta \hat{n})$ gives the effective length (scalar) $\perp$ to the magnetic field with the direction $(\vec{F})$ of net force on complete current carrying wire. So, we can write simplified expression in following way–

$$\vec{F} = (i\ B\ l\ )\hat{F}$$

or $$\boxed{\vec{F} = i\ \vec{l} \times \vec{B}}$$

where $\vec{l}$ is the vector joining initial point (i) to final point (f) of the given wire.

From the above discussion, we can say that any current carrying closed loop present in a uniform field, will experience no force as the initial (i) and final point (f) meet at the same location though it may experience torque.

Q. **Find the magnetic force m the given random shaped wire present in a uniform magnetic field B as shown in figure.**



Sol. **Method–1**

$\vec{F} = i\int d\vec{l} \times \vec{B}$

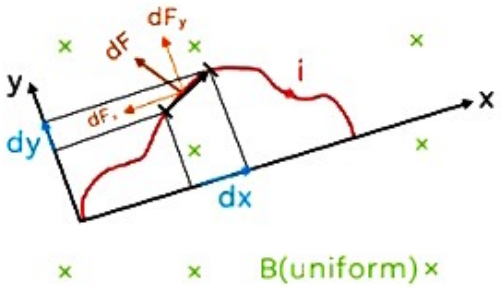
$\vec{F} = i\int (dx\hat{i} + dy\hat{j}) \times B(-\hat{k})$

$\vec{F} = i\int (dx.B)\hat{j} + i\int dy.B(-\hat{i})$

$\vec{F} = iB\int (dx)\hat{j} - iB\int (dy)\hat{i}$

$\vec{F} = iB(l)\hat{j} - iB(0)\hat{i}$

$\boxed{\vec{F} = (ilB)\hat{j}}$



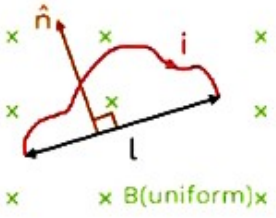
Method-2

The effective length between the end point is l and the angle b/w $\vec{l}$ and $\vec{B}$ is 90°.

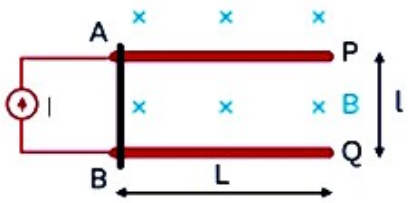
$\therefore \quad \vec{F} = i\vec{l} \times \vec{B}$

$\vec{F} = ilB\sin 90°\hat{n}$

$\vec{F} = ilB\hat{n}$

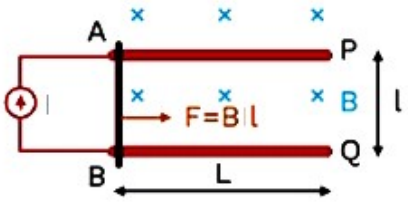


Q. **A rod AB of mass m is gently placed on two smooth rails P and Q in a uniform magnetic induction B as shown in figure. The rails are connected to a constant current source of current I. Find the speed attained by rod AB when it leaves off the other end of rails.**



Sol. Acceleration due to magnetic force is –

$$a = \frac{BIl}{m}$$



Final speed attained by the rod can be found using equation of motion,

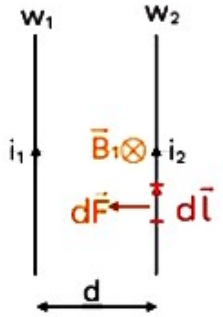
$$v^2 = u^2 + 2aL$$

$$\boxed{v = \sqrt{\frac{2BIlL}{m}}}$$

Force Between Parallel Current Carrying Wires

Consider two long wires W_1 and W_2 kept parallel as shown in figure. Consider a small element $d\vec{l}$ of the wire W_2. The magnetic field at $d\vec{l}$ due to the wire W_1 is :–

$$\vec{B}_1 = \frac{\mu_0 i_1}{2\pi d} \otimes \qquad \text{...(i)}$$



The field due to wire W_2 at the location of $d\vec{l}$ is zero. Thus, equation (i) gives the net field at $d\vec{l}$. The magnetic force at the element $d\vec{l}$ is –

$$d\vec{F} = i_2 d\vec{l} \times \vec{B}_1$$

$$dF = i_2 dl \frac{\mu_0 i_1}{2\pi d}$$

The force per unit length of the wire W_2 due to the wire W_1 is –

$$\boxed{\frac{dF}{dl} = \frac{\mu_0 i_1 i_2}{2\pi d}} \qquad \text{....(ii)}$$

If we take an element $d\vec{l}$ on the wire W_1 and calculate the magnetic force per unit length of wire W_1 due to W_2, it is again given by equation (2).

For directions, we can say – If the parallel wires carry current in same directions, the wires attract each other and vice-versa.

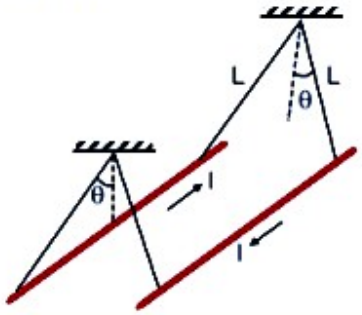
Definition of Ampere: In above case, if, $i_1 = i_2 = 1$ A and d = 1 m then, from equation (2)–

$$\frac{dF}{dl} = 2 \times 10^{-7} \text{ N/m}$$

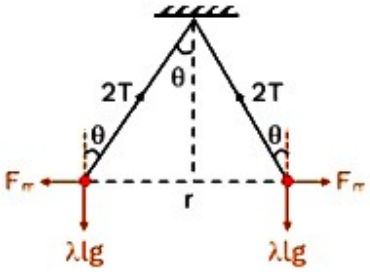
This is used to formally define the unit 'ampere' of electric current and defined as–

"If two parallel, long wires kept 1m apart in vacuum, carry equal currents in the same direction and there is a force of attraction of 2×10^{-7} N/m of each wire, the current in each wire is said to be 1 ampere."

Q. **Two long parallel wires carry current of equal magnitude but in opposite direction. These wires are suspended from ceiling by four strings of same length L as shown. The mass per unit length of the wires is λ. Determine the value of θ assuming it to be small.**



Sol.



FBD (cross-sectional view)

Let us consider tension in each string in T, length of each rod is l and distance between rods is r.

In equilibrium,

$2T\cos\theta = \lambda lg$ .....(1)

$2T\sin\theta = F_m = \left(\frac{dF}{dl}\right)l$

$2T\sin\theta = \frac{\mu_0 I^2}{2\pi r}l$ .....(2)

From (1) and (2),

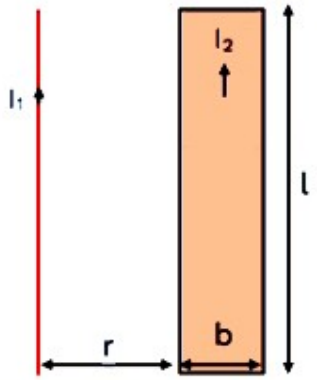
$$\tan\theta = \frac{\mu_0 I^2}{2\pi\lambda g r} \Rightarrow \tan\theta = \frac{\mu_0 I^2}{4\pi\lambda g L\left(\frac{r/2}{L}\right)}$$

$$\Rightarrow \tan\theta = \frac{\mu_0 I^2}{4\pi\lambda g L\tan\theta} \Rightarrow \tan^2\theta = \frac{\mu_0 I^2}{4\pi\lambda g L}$$

For small angle,

$$\theta^2 \approx \frac{\mu_0 I^2}{4\pi\lambda g L} \Rightarrow \boxed{\theta = \sqrt{\frac{\mu_0 I^2}{4\pi\lambda g L}}}$$

Q. **Figure shows a long straight current carrying wire with a current I_1 and a thin strip of width b and length l is placed parallel to it with a current I_2 as shown. Find the magnetic force of interaction between currents I_1 and I_2.**



Sol. Current in elemental wire,

$$dI_2 = \frac{I_2}{b}.dx$$

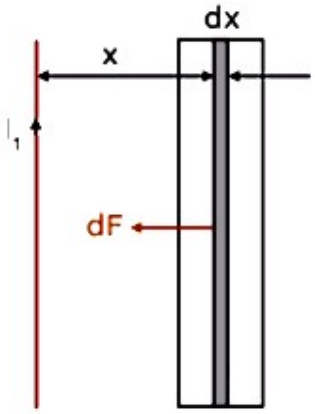
Force b/w I_1 and dI_2 is –

$$dF = \frac{\mu_0 I_1 (dI_2) l}{2\pi x}$$

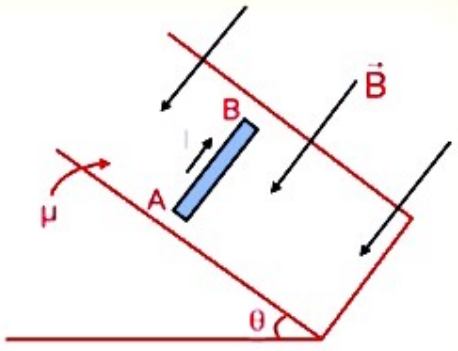
Net force on strip –

$$F = \int dF = \frac{\mu_0 I_1 I_2 l}{2\pi b} \int_r^{r+b} \frac{dx}{x}$$

$$\boxed{F = \frac{\mu_0 I_1 I_2 l}{2\pi b} \ln\left(1 + \frac{b}{r}\right)}$$

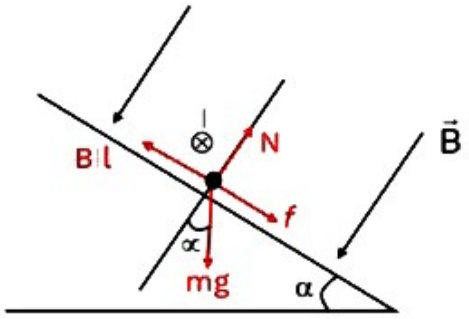


Q. **Figure shows a wire AB of mass m, length l placed on a rough inclined plane of inclination α, it carries a current I. Find the minimum magnitude of magnetic induction ($\perp$ to the plane) required to slide the wire up the inclined plane.**



Sol.

FBD of rod (in cross-sectional view) is shown in the figure.



For wire to slide up the incline,

$BIl \geq f + mg\sin\alpha$

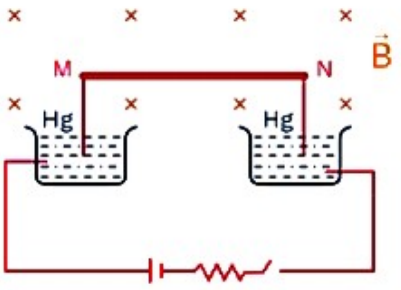
$BIl \geq \mu N + mg\sin\alpha$

$BIl \geq \mu mg\cos\alpha + mg\sin\alpha$ ...(As $N=mg\cos\alpha$)

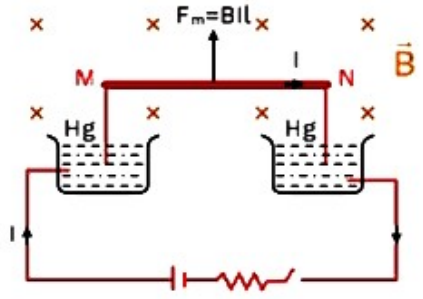
$$B \geq \frac{mg(\mu\cos\alpha + \sin\alpha)}{Il}$$

$$\therefore \quad \boxed{B_{min} = \frac{mg(\mu\cos\alpha + \sin\alpha)}{Il}}$$

Q. **Figure shows a horizontal wire MN of length l and mass m is placed in a uniform magnetic field B. The ends of wire are bent and dipped in two bowls containing Hg which is connected to an external circuit as shown. As key is pressed, a charge q flows through the circuit for a short time δt and the wire MN looses the contact with the mercury. Find the maximum height above initial level the wire MN will jump.**



Sol. The wire moves up within a short time δt, i.e. magnetic force F_m will be an impulsive force (much greater than mg). The impulse transferred by magnetic force to the wire is,



$J = F_m \delta t = mu$

$BIl\ \delta t = mu$

$Bql = mu$

$$u = \frac{Bql}{m}$$

where u = initial velocity of wire (when it looses the contact with Hg).

Maximum height attained by wire MN–

$$h_{max} = \frac{u^2}{2g}$$

$$\Rightarrow \boxed{h_{max} = \frac{B^2q^2l^2}{2m^2g}}$$

Motion of a Charged Particle in Combined Electric and Magnetic Field

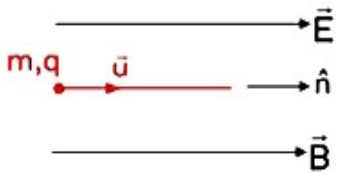
Suppose a charged particle moving with velocity $\vec{v}$ (at some instant) in an electromagnetic field, then the Lorentz force on the charged particle at that instant will be

$$\vec{F} = \vec{F}_{el.} + \vec{F}_{mag.}$$

$$\vec{F} = q\vec{E} + q(\vec{v} \times \vec{B})$$

The force $\vec{F}$ and velocity $\vec{v}$ are inter-dependent, i.e., $\vec{F}$ will change the $\vec{v}$ and the value of new $\vec{v}$ will change the value of $\vec{F}$. So, the motion of charged particle will not be simple. It actually varies from case to case. Let us discuss some common cases.

Case 1: When $\vec{E}, \vec{B}$ and $\vec{u}$ all are parallel:



Suppose

$$\vec{u} = u\hat{n} \text{ (at } t = 0);$$

$$\vec{B} = B_0\hat{n};$$

$$\vec{E} = E_0\hat{n};$$

and both $\vec{B}$ and $\vec{E}$ are constant and uniform.
The initial force on q is-

$$\vec{F} = q\vec{E} + q(\vec{u} \times \vec{B}) = qE_0\hat{n} + \vec{0}$$

$$\vec{a} = \frac{qE_0}{m}\hat{n}$$

At any time t,

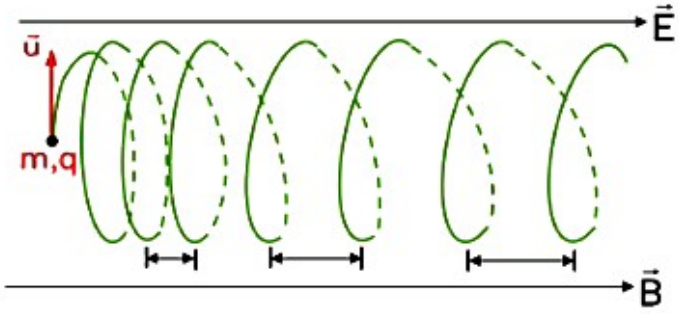
$$\vec{v} = \vec{u} + \vec{a}t$$

$$\Rightarrow \vec{v} = u\hat{n} + \frac{qE_0t}{m}\hat{n}$$

$$\vec{v} = \left(u + \frac{qE_0t}{m}\right)\hat{n}$$

The charged particle will move along a straight line path and it will be a uniformly accelerated motion.

Case 2: When $\vec{E}$ is $\parallel$ to $\vec{B}$ and $\vec{v}$ is $\perp$ to both fields :



Suppose

$\vec{E} = E_0\hat{i}$;

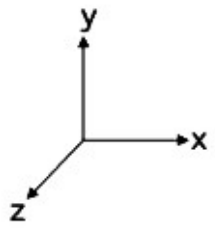
$\vec{B} = B_0\hat{i}$;

$\vec{v} = u_0\hat{j}$, (at t = 0) ;

and both $\vec{E}$ and $\vec{B}$ are uniform and constant.

$\because$ $\vec{B}$ is along x-axis,

$\therefore$ Magnetic force $\vec{F}_m$ will always be $\perp$ to x-axis and only electric force $\vec{F}_{el}$ will be along x-axis.



$$\vec{F}_{el} = qE_0\hat{i} \quad \Rightarrow \vec{a}_{el} = \frac{qE_0}{m}\hat{i} = \text{constant}$$

$$\therefore \quad v_x = u_x + a_x t \Rightarrow v_x = 0 + \frac{qE_0}{m}t$$

$$\Rightarrow \quad v_x = \frac{qE_0}{m}t \qquad ...(1)$$

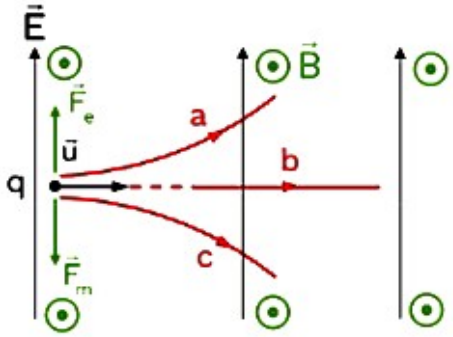
$$\text{Also, } \Delta x = u_x t + \frac{1}{2}a_x t^2 \quad \Rightarrow \Delta x = 0 + \frac{qE_0 t^2}{2m}$$

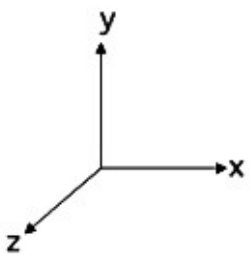
$$\Delta x = \frac{qE_0}{2m}t^2 \qquad ...(2)$$

Due to magnetic force, particle will attempt to execute circular motion in y-z plane with radius, $R = \frac{mu}{qB}$ and time period, $T = \frac{2\pi m}{qB}$

Now we can say that motion of the particle will be non-uniform helix with radius R, time period T and increasing pitch (as we move along x-axis).

Case 3: When all $\vec{B}$, $\vec{v}$ and $\vec{E}$ are mutually perpendicular :





Assume q > 0, electric force on charge,

$\vec{F}_e = qE_0\hat{j}$ or $F_e = qE_0$

Magnetic force on charge,

$\vec{F}_m = quB_0(-\hat{j})$ or $F_m = quB_0$

Situation (a): If $F_e > F_m \Rightarrow qE_0 > quB_0 \Rightarrow E_0 > uB_0$, particle will move along parabolic path (a).

Situation (b): If $F_e = F_m \Rightarrow E_0 = uB_0 \Rightarrow u = \frac{E_0}{B_0}$, particle will move along straight line path (b) with no acceleration. We call this situation as "velocity selector".

Situation (c): If $F_e < F_m \Rightarrow E_0 < uB_0$, particle will move along parabolic path (c).

Q. **A +ve charge q with mass m is released from origin in a coordinate system where a uniform electric field $\vec{E} = E_0\hat{j}$ and magnetic field $\vec{B} = B_0\hat{k}$ exist. Find the displacement of the particle along y-axis when its velocity becomes perpendicular to electric field.**

Sol. Acceleration at point P,

$$a_x = \frac{q\upsilon B_0 \sin\theta}{m}$$

$$a_x = \frac{q\upsilon_y B_0}{m} \quad ...(1)$$

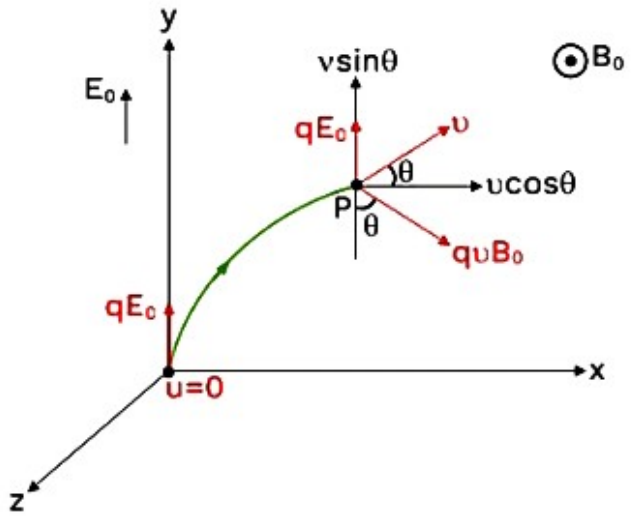
$$a_y = \frac{qE_0 - q\upsilon B_0 \cos\theta}{m}$$

$$a_y = \frac{qE_0 - q\upsilon_x B_0}{m} \quad ...(2)$$

from equation (2),

$$\upsilon_x = \frac{qE_0 - ma_y}{qB_0}$$

$$\upsilon_x = \left(\frac{qE_0}{m} - \frac{d\upsilon_y}{dt}\right)\cdot\frac{m}{qB_0}$$



differentiate the equation w.r.t. t,

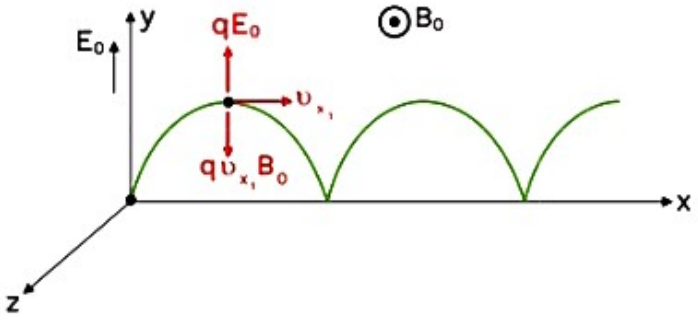
$$\frac{d\upsilon_x}{dt} = -\frac{m}{qB_0}\frac{d^2\upsilon_y}{dt^2}$$

from equation (1),

$$a_x = \frac{d\upsilon_x}{dt} = \frac{q\upsilon_y B_0}{m} = -\frac{m}{qB_0}\frac{d^2\upsilon_y}{dt^2}$$

$$\boxed{\frac{d^2\upsilon_y}{dt^2} + \left(\frac{qB_0}{m}\right)^2 \upsilon_y = 0}$$ (SHM equation in υ_y)

This equation suggests that the path of charged particle will be cycloid (as shown below).



At top point, $\vec{\upsilon}_{x_1}$ is $\perp$ to $\vec{E}$.

$\because$ Here, initial point and top point are the extreme points of SHM.

$$\therefore \quad a_{initial} = a_{top}$$

$$\Rightarrow \quad \frac{qE_0}{m} = \frac{q\upsilon_{x_1}B_0 - qE_0}{m} \quad \Rightarrow \upsilon_{x_1} = \frac{2E_0}{B_0}$$

Use work-energy theorem between initial and top point-

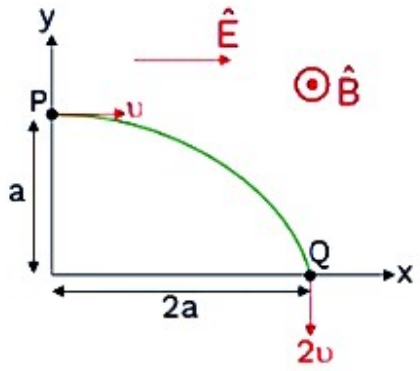
$W_{el} + W_m = \Delta K$

$$qE_0(y_{max}) + 0 = \frac{1}{2}m(\upsilon_{x_1})^2$$

$$qE_0\, y_{max} = \frac{1}{2}m\left(\frac{4E_0^2}{B_0^2}\right)$$

$$\boxed{y_{max} = \frac{2mE_0}{qB_0^2}}$$

Q. **Figure shows the trajectory of a charged particle (q, m) placed in a uniform electric and magnetic fields. The particle enters in region at point P with speed υ and leaves at point Q with speed 2υ. Find the power supplied by the electric and magnetic field to the charge at point P.**



Sol. $\because$ W.D. by magnetic field = 0

$\therefore$ Power supplied by magnetic field = 0 at any instant or $\boxed{P_m = 0}$

Suppose the value of electric field = E_0

Now, apply Work-Energy theorem between P and Q,

$W_{el} + W_m = \Delta K$

$$qE_0(2a) + 0 = \frac{1}{2}m(2\upsilon)^2 - \frac{1}{2}m\upsilon^2$$

$$E_0 = \frac{3m\upsilon^2}{4qa}$$

$\therefore$ Power supplied by $\vec{E}$ at point P is-

$P_e = F\upsilon = qE_0\upsilon$

$$\boxed{P_e = \frac{3m\upsilon^3}{4a}}$$

Q. **A particle (m, q) is released from the origin in a region in which electric and magnetic fields are given by-**
$\vec{B} = -B_0\hat{j}$ **and** $\vec{E} = E_0\hat{k}$

Find the speed of the particle as a function of its z-coordinate.

Sol. Suppose, speed at z-coordinate = υ

Apply work-energy theorem from z = 0 to z = z,

$W_{el} + W_m = \Delta K$

$F_e(z - 0) + 0 = \frac{1}{2}m\upsilon^2 - 0$

$qE_0z = \frac{1}{2}m\upsilon^2$

$\Rightarrow \boxed{\upsilon = \sqrt{\frac{2qE_0z}{m}}}$

Q. **A proton goes undeflected in a crossed electric and magnetic field (the fields are perpendicular to each other) at a speed of 2.0×10^5 m/s. The velocity is perpendicular to both the fields. When the electric field is switched off, the proton moves along a circle of radius 4.0 cm. Find the magnitudes of the electric and the magnetic fields. Take the mass of the proton = 1.6×10^{-27} kg.**

Sol. When there is no deflection, $\vec{F}_{net} = \vec{0}$

or $F_e = F_m$

$eE = e\upsilon B \Rightarrow E = \upsilon B$

$\Rightarrow E = 2 \times 10^5 B$ ...(1)

When E = 0 and B ≠ 0,

$R = \frac{m\upsilon}{eB}$

$\Rightarrow B = \frac{m\upsilon}{eR} = \frac{1.6 \times 10^{-27} \times 2 \times 10^5}{1.6 \times 10^{-19} \times 4 \times 10^{-2}}$

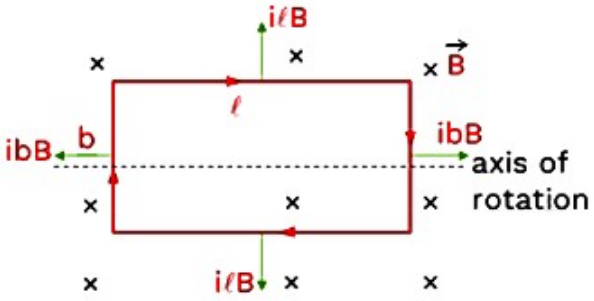
$\Rightarrow \boxed{B = 0.05\ T}$

From equation (1),

$\boxed{E = 10^4\ V/m}$

Torque on a Current Loop Present in Uniform Magnetic Field

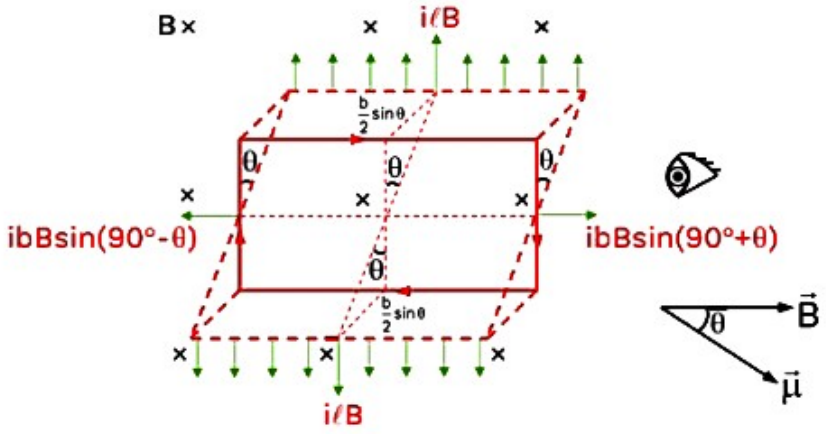
Case 1:



$\left(\sum \vec{F}\right)_{loop} = 0 \Rightarrow$ Loop will not go anywhere as net force on it is zero.

$\left(\sum_{loop} \vec{\tau}\right)_{axis} = 0 \Rightarrow$ Also torque about any axis is zero as separation between line of action of forces is zero.

Case 2:



$\sum \vec{F}_{loop} = 0 \Rightarrow$ Loop will not go anywhere.

$$\left(\sum_{loop} \vec{\tau}\right)_{axis} = i\ell B\frac{b}{2}\sin\theta + i\ell B\frac{b}{2}\sin\theta + 0 + 0$$

$$\left(\sum_{loop} \vec{\tau}\right)_{axis} = i\ell bB\sin\theta \quad (\text{Anticlockwise}) = i\vec{A}\times\vec{B} = \vec{\mu}_{loop}\times\vec{B}$$

$$\vec{\tau} = -(iAB\sin\theta)\hat{\theta}$$

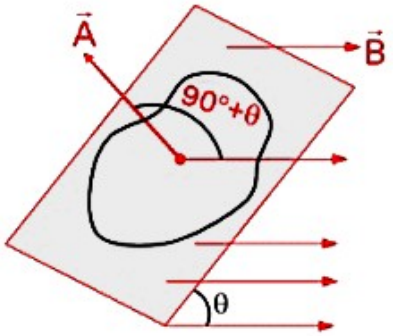
It is kind of restoring torque for N turns (or coil),

$$\vec{\tau} = Ni\vec{A} \times \vec{B} = \vec{\mu}_{coil} \times \vec{B}$$

where $\vec{\mu}_{coil}$ = magnetic moment or magnetic dipole moment of coil

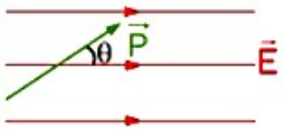
Here, $\vec{A} = (\ell\vec{b})$ = Area vector of loop and it is along the motion of a right-hand screw and screw should be rotated along the direction of current.

Note: The above equations for torque are valid for planar loops of any shape present in uniform magnetic field. This torque concept is used in ammeter, galvanometer etc.



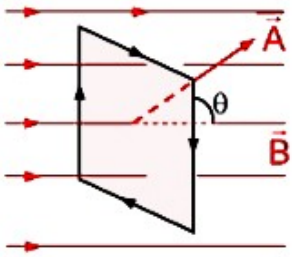
$$= i\vec{A} \times \vec{B} = i\,A\,B\,\sin(90° + \theta)$$

Positional / Potential / Interaction Energy of a Current Carrying Loop in a Uniform Magnetic Field



$U_\theta = -\vec{P}.\vec{E}$

$\vec{\tau} = \vec{P} \times \vec{E}$



$U(\theta) = -\vec{\mu}.\vec{B}$

$\vec{\tau} = \vec{\mu} \times \vec{B}$

Work Done in changing the orientation of a current loop in uniform Magnetic field :



$U_1 = -\mu B \cos\theta_i$ $\qquad$ $U_2 = -\mu B \cos\theta_f$

$W.D._{ext} = \Delta U = U_f - U_i = iAB(\cos\theta_i - \cos\theta_f)$

$= \left[(\vec{B}.\vec{A})_i - (\vec{B}.\vec{A})_f\right] = i(\phi_i - \phi_f)$

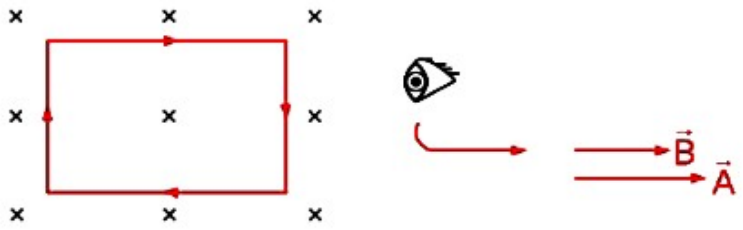
$= -i(\phi_f - \phi_i) = -i\Delta\phi$

$W.D._{ext} = -i\Delta\phi$

where $\phi = \vec{B}.\vec{A}$ is termed as magnetic flux of $\vec{B}$ passing through the coil.

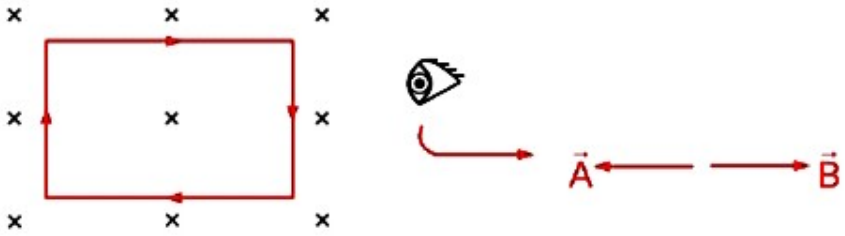
Stable & Unstable equation of current loops in uniform Magnetic Field:

Case 1:



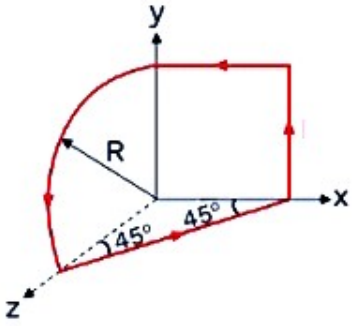
Here $\vec{A}$ & $\vec{B}$ are parallel or $\theta = 0°$, $\tau = 0$ but $U = -\mu B$ = minimum, system is in stable equilibrium.

Case 2:

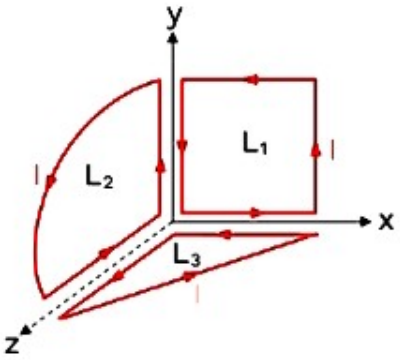


Here $\vec{A}$ & $\vec{B}$ are antiparallel or $\theta = 180°$, $\tau = 0$ but $v = +\mu B = IAB$ = maximum, system is in unstable equilibrium.

Q. **Find the magnetic moment of the current carrying loop shown in figure.**



Sol.

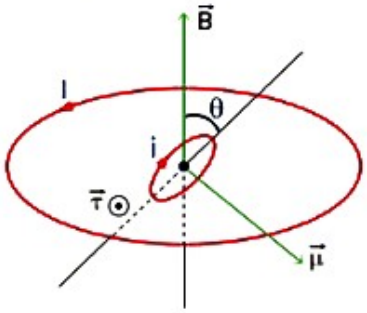


The whole loop can be splitted in the following 3 simple loops,

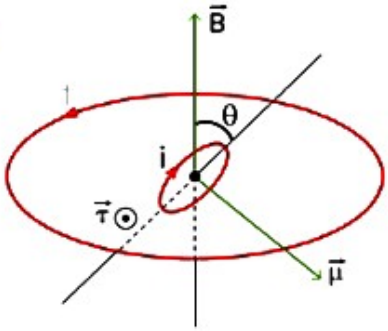
$$\vec{M}_{loop} = \vec{M}_1 + \vec{M}_2 + \vec{M}_3$$

$$\vec{M}_{loop} = I(R^2)\hat{k} + I\left(\frac{\pi R^2}{4}\right)\hat{i} + I\left(\frac{R^2}{2}\right)\hat{j}$$

Q. **Figure shows a large horizontal coil of radius R carrying a current I. Another small coil of radius r (r < < R) carrying a current i and N turns is placed at the centre with its plane at an angle θ from the axis. Find the torque experienced by the smaller coil for the given situation.**



Sol.



Magnetic moment of small coil,

$\mu = i \pi r^2 N = N i \pi r^2$

Magnetic field due to large coil at its centre, $B = \frac{\mu_0 I}{2R}$

Angle between $\vec{B}$ and $\vec{\mu}$ $= \frac{\pi}{2} + \theta$,

Torque on smaller coil,

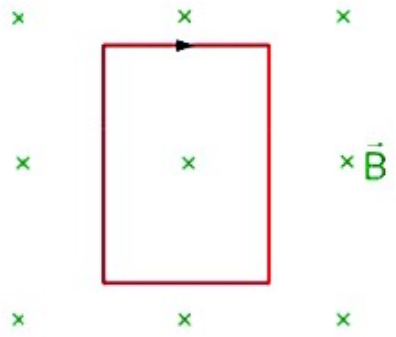
$$\vec{\tau} = \vec{\mu} \times \vec{B}$$

$$\vec{\tau} = \mu B \sin\left(\frac{\pi}{2} + \theta\right)$$

$$\tau = N i \pi r^2 \left(\frac{\mu_0 I}{2R}\right) \cos\theta$$

$$\boxed{\tau = \frac{\mu_0 i I \pi r^2 N}{2R} \cos\theta}$$

Q. **In figure shown a coil area A and N turns with a current I is placed in a uniform field B. Find the work required to pull this coil out from magnetic field.**



Sol. Work done in this process-

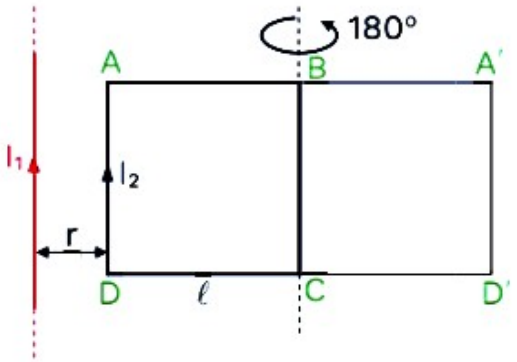
$W = -I\,\Delta\phi \Rightarrow W = I(\phi_1 - \phi_2)$

Initial flux through coil $\phi_1 = BAN$

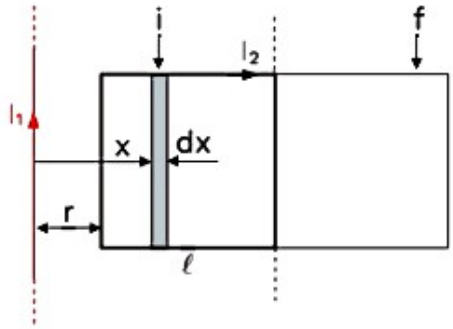
Final flux through coil $\phi_2 = 0$

Work done, $W = I(BAN - 0) \Rightarrow \boxed{W = BINA}$

Q. **A square coil of edge ℓ carrying a current I_2 is placed near to a long straight wire carrying current I_1 as shown. Find work required to rotate the coil ABCD about the axis along edge BC by 180° as shown.**



Sol.



W.D. in the rotating process,

$$W = I_2(\phi_i - \phi_f) \qquad ...(1)$$

Flux in initial position,

$$\phi_i = \int B_1 dA = \int_r^{r+\ell} \frac{\mu_0 I_1}{2\pi x}.\ell dx$$

$$\phi_i = \frac{\mu_0 I_1 \ell}{2\pi}\left[\ell n x\right]_r^{r+\ell}$$

$$\phi_i = \frac{\mu_0 I_1 \ell}{2\pi}\ell n\left(\frac{r+\ell}{r}\right) \qquad ...(2)$$

Flux in final position,

$$\phi_f = -\int B_1 dA = -\int_{r+\ell}^{r+2\ell} \frac{\mu_0 I}{2\pi x}.\ell dx = -\frac{\mu_0 I_1 \ell}{2\pi}\left[\ell n x\right]_{r+\ell}^{r+2\ell}$$

$$\phi_f = -\frac{\mu_0 I_1 \ell}{2\pi}\ell n\left(\frac{r+2\ell}{r+\ell}\right) \qquad ...(3)$$

Using all equations,

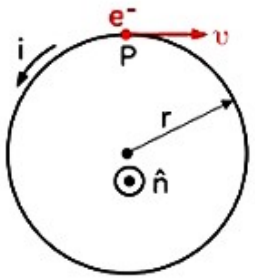
$$W = I_2\frac{\mu_0 I_1 \ell}{2\pi}\left[\ell n\left(\frac{r+\ell}{r}\right) + \ell n\left(\frac{r+2\ell}{r+\ell}\right)\right]$$

$$\boxed{w = \frac{\mu_0 I_1 I_2 \ell}{2\pi}\ell n\left(\frac{r+2\ell}{r}\right)}$$

Magnetic Dipole Moment of a Revolving Electron

Consider a point P on the path of the electron. In one revolution of the electron, a charge e crosses the point P. Therefore, the equivalent electric current-

$$i = \frac{e}{T} = \frac{e}{\left(\frac{2\pi r}{\upsilon}\right)} = \frac{e\upsilon}{2\pi r}$$



The magnetic moment of the circulating electron-

$$\mu = iA = \frac{e\upsilon}{2\pi r}.\pi r^2$$

$$\mu = \frac{e\upsilon r}{2}$$

or $\boxed{\vec{\mu} = \frac{e\upsilon r}{2}\hat{n}}$ ...(1)

The angular momentum of revolving electron is-

$$\vec{L} = m\vec{r} \times \vec{\upsilon} \text{ or } \vec{L} = m\upsilon r\,(-\hat{n}) \quad ...(2)$$

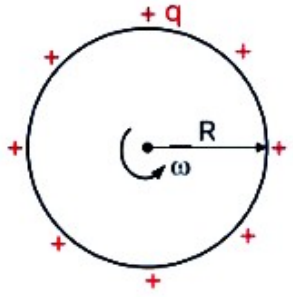
Now,

$$\boxed{\frac{\mu}{L} = \frac{-e}{2m}}$$

'–' sign shows that the direction of angular momentum is opposite to that of the magnetic moment.

The term $\frac{\mu}{L}$ is also known as "Gyromagnetic ratio".

Q. Find the magnetic moment of a charged ring (q, m, R) rotating with angular velocity ω about its axis. Also find the ratio $\frac{\mu}{L}$.



Sol. Equivalent current,

$$i = \frac{q}{T} = \frac{q\omega}{2\pi}$$

Magnetic moment,

$$\mu = \pi R^2 \cdot \frac{q\omega}{2\pi} = \frac{q\omega R^2}{2}$$

Angular momentum,

$$L = I\omega = mR^2\omega$$

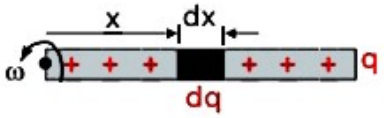
$$\therefore \quad \frac{\mu}{L} = \frac{q}{2m} = \text{constant for all } \omega \text{ and } R.$$

Q. **Find the ratio of magnetic moment of a charged rod (m, ℓ, q) rotating with ω about its one end, to its angular momentum. Assume charge and mass are uniformly distributed.**

Sol.

$$L = I\omega$$

$$L = \frac{m\ell^2\omega}{3} \qquad ...(1)$$



Equivalent elemental current due to 'dx' section,

$$d\mu = (di)A = \frac{dq}{T}\pi x^2 = \frac{q}{\ell}.dx.\frac{1}{\left(\frac{2\pi}{\omega}\right)}\pi x^2$$

$$d\mu = \frac{q\omega x^2 dx}{2\ell}$$

$$\therefore \quad \mu = \frac{q\omega}{2\ell}\int_0^{\ell} x^2 dx \qquad \Rightarrow \quad \mu = \frac{q\omega}{2\ell}\left(\frac{\ell^3}{3}\right)$$

$$\Rightarrow \quad \mu = \frac{q\omega\ell^2}{6} \qquad ...(2)$$

From (1) and (2),

$$\frac{\mu}{L} = \frac{q}{2m} = \text{constant}$$

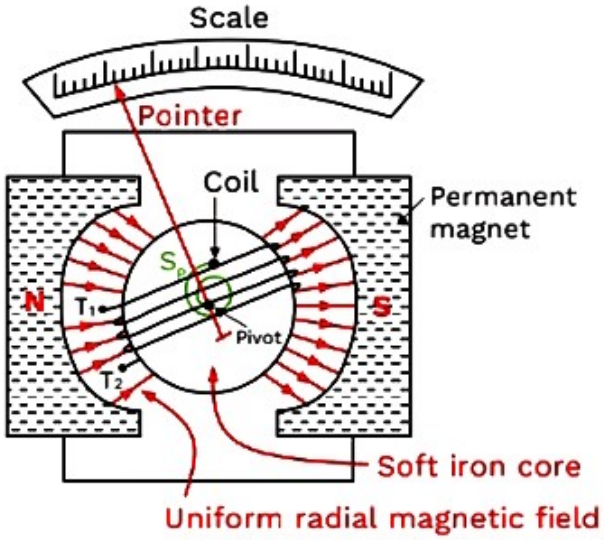
If q is –ve, the ratio will be negative.

Note: All rotating objects (ring, rod, disc, spherical shell, solid sphere, etc.) having the uniform mass and charge distribution, their $\frac{\mu}{L}$ ratio is always equal to $\frac{q}{2m}$.

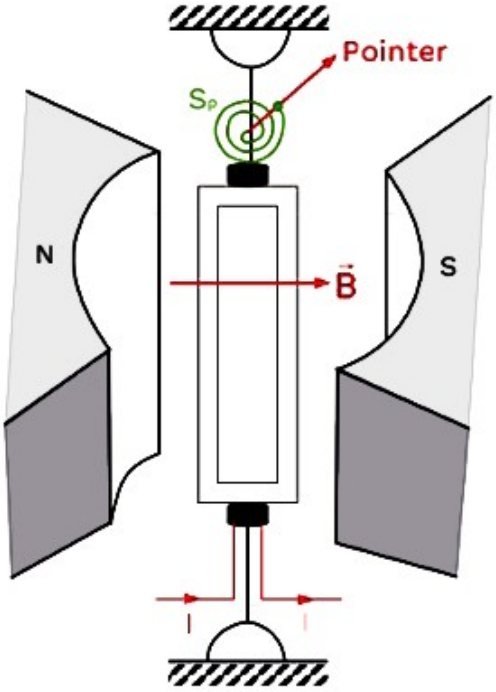
Moving-Coil Galvanometer

A galvanometer is a device to detect current in a circuit, i.e., it can work as an ammeter as well as a voltmeter.

Principle: A current carrying coil placed in a magnetic field experiences a torque, the magnitude of which depends on the strength of current.



Top view of a pivoted type galvanometer



(Front view of galvanometer)

A Weston (pivoted type) galvanometer consists of a rectangular coil of fine insulted copper wire wound on a light non-magnetic metallic (aluminium) frame. The two ends of the axle of this frame are pivoted between two jewelled bearings. The motion of the coil is controlled by a pair of hair springs of phosphor bronze. The springs provide the restoring torque and serve as current leads. A light aluminium pointer attached to the coil measures its deflection on a suitable scale.

Working: When a current flows through the coil, a torque act on it. This torque is given by-

$\tau = NIAB\sin\theta$

$\because$ The field is radial by design, we will take, $\sin\theta = 1$

$\therefore \quad \tau = NIAB \qquad ...(1)$

where I = Current flowing through the coil,

A = Area of the coil,

N = Number of turns in the coil,

B = Magnetic field due to magnet at the plane of coil.

The magnetic torque NIAB tends to rotate the coil. A spring S_p provides a counter torque $K\phi$ that balances the magnetic torque NIAB, resulting in a steady angular deflection ϕ.

In equilibrium-

Restoring torque = Deflecting torque

$K\phi = NIAB$

$$\boxed{\phi = \left(\frac{NAB}{K}\right)I} \qquad ...(2)$$

where K = torsional constant of the spring or resulting torque per unit twist,

ϕ = angular deflection in steady state.

The quantity $\left(\frac{NAB}{K}\right)$ is a constant for a given galvanometer and therefore-

$\phi \propto I$

This shows that the galvanometer should have a linear scale to indicate equal steps in current.

Current Sensitivity :

$$I_s = \frac{\phi}{I} = \frac{NBA}{K}$$

It is defined as deflection in galvanometer per unit current.

Voltage Sensitivity :

$$V_s = \frac{\phi}{V} = \frac{\phi}{IR} = \frac{NBA}{KR}$$

$$V_s = \frac{I_s}{R}$$

It is defined as deflection in galvanometer per unit potential difference (applied across galvanometer).

Q. **A rectangular coil of area 5×10^{-4} m² and 60 turns is pivoted about one of its vertical sides. The coil is in a radial horizontal field of 90 G. What is the torsional constant of the hair-springs connected to the coil if a current of 0.2 mA produces an angular deflection of 18°?**

Sol. $$K = \frac{NIBA}{\phi} = \frac{60 \times 0.2 \times 10^{-3} \times 90 \times 10^{-4} \times 5 \times 10^{-4}}{18}$$

$K = 3 \times 10^{-9}$ N-m/degree

Q. **A galvanometer needs 50 mV for a full-scale deflection of 50 divisions. Find its voltage sensitivity. What must be its resistance if its current sensitivity is 1 division/μA?**

Sol. $$V_s = \frac{\phi}{V} = \frac{50\,\text{div}}{50\,\text{mV}} = 10^3 \text{ div./V}$$

Resistance of galvanometer,

$$R_g = \frac{I_s}{V_s} = \frac{1\text{div.} / \mu\text{A}}{10^3 \text{ div.} / \text{V}} = \frac{1}{10^3 \times 10^{-6}}\Omega$$

$R_g = 1000\ \Omega$

Q. **A MCG has the following particulars: Number of turns, N = 24 ; Area of coil, $A = 20 \times 10^{-3}$ m² ; magnetic field strength, B = 0.20 T ; resistance of the coil, R = 14 Ω.**

(i) Indicate a simple way to increase the current sensitivity of the meter by 25% (it is not easy to change A or B).

(ii) If in doing so, the resistance of the coil increases to 21 Ω, is the voltage sensitivity of the modified meter greater or less than the original meter?

Sol. (i) $I_s = \frac{\phi}{I} = \frac{NBA}{K}$,

Since it is easier to change N as compare to K, A or B. So, the I_s can be increased by increasing N by 25%, i.e., increase N from 24 to 30.

(ii) $V_s = \frac{NBA}{KR}$. As K, A and B are same in two case, we need to compare only $\frac{N}{R}$ ratio.

For original meter, $\frac{N}{R} = \frac{24}{14} = 1.7$

For modified meter, $\frac{N}{R} = \frac{30}{21} = 1.4$

Thus, the modified meter has less V_s than its original meter but it has gained current sensitivity.